

Versionierungsunterstützung medizinischer Terminologien im Kontext semantischer Mappings

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Studiengang: Informatik

Projekt: „Miracum Mapper 2.1“

Betreut von: PD Dr. Dennis Toddenroth, Noemi Deppenwiese M. Sc.

Motivation

- **Terminologie / Codesystem:** besteht aus Konzepten (z.B. LOINC, SNOMED CT, ICD-10-GM)
 - **Mappen:** Konzepte aus einer Terminologie mit Konzepten aus einer anderen Terminologie in Beziehung setzen
 - Ziel: Semantische Interoperabilität
 - Mappen ist ein sehr zeitaufwendiger Prozess, erfordert Expertenwissen
 - Unterstützung des Mappingprozesses durch ein Tool (Mapper)
- **Miracum Mapper 2.0** (entwickelt von Nina Mücke und Henrik Herzig)



MiracumMapper

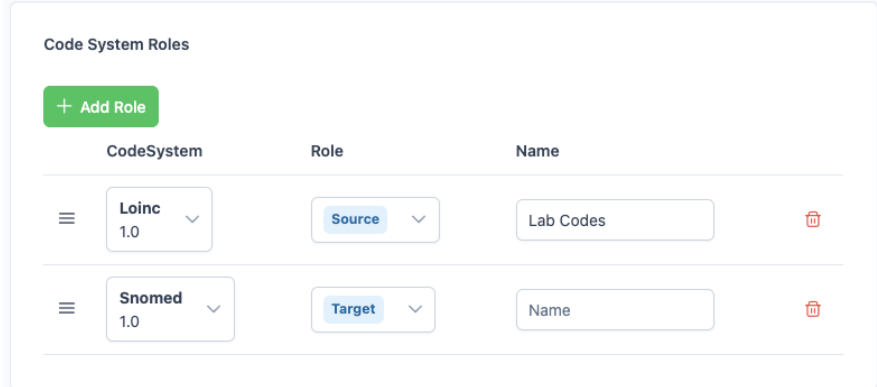
- **Terminologien** sind nicht „fest“ sondern **verändern sich regelmäßig**
 - **Neue Versionen** mit neuen Konzepten und Änderungen
- Frequenz der neuen Versionen abhängig von der Terminologie:
 - ICD-10-GM: 1 mal jährlich
 - LOINC: ~ 2 mal jährlich
 - SNOMED CT: 1 mal monatlich
- Beispiel: LOINC Version 2.78 (06.08.2024)
 - Neu: 105910-4 – SARS-CoV-2 (COVID-19) IgG Ab [Units/volume] in Specimen
 - Status geändert: 101969-4 – Facial Expression **Active** → **Deprecated**
 - Meaning geändert: 104004-7 – US for pregnancy in third trimester → Ultrasound for pregnancy in third trimester

- **Problem:** Miracum Mapper 2.0 unterstützt neue Versionen nicht direkt

Ziele

- **Speicherung der Änderungen** zwischen zwei Versionen
- **Versionsverwaltung** innerhalb der Terminologien / Codesysteme
- **Migration eines Projekts** auf eine neue Version
 - Anzeige der Änderungen zwischen zwei Versionen
 - Manuelles um-mappen bei gelöschten Konzepten

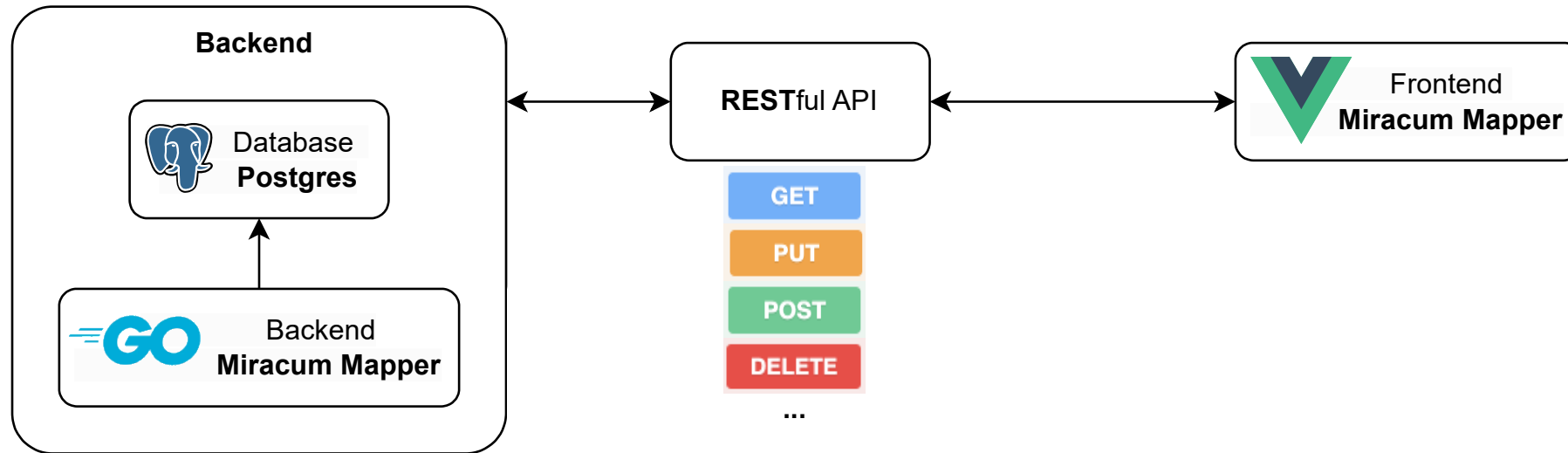
→ **Miracum Mapper 2.1**



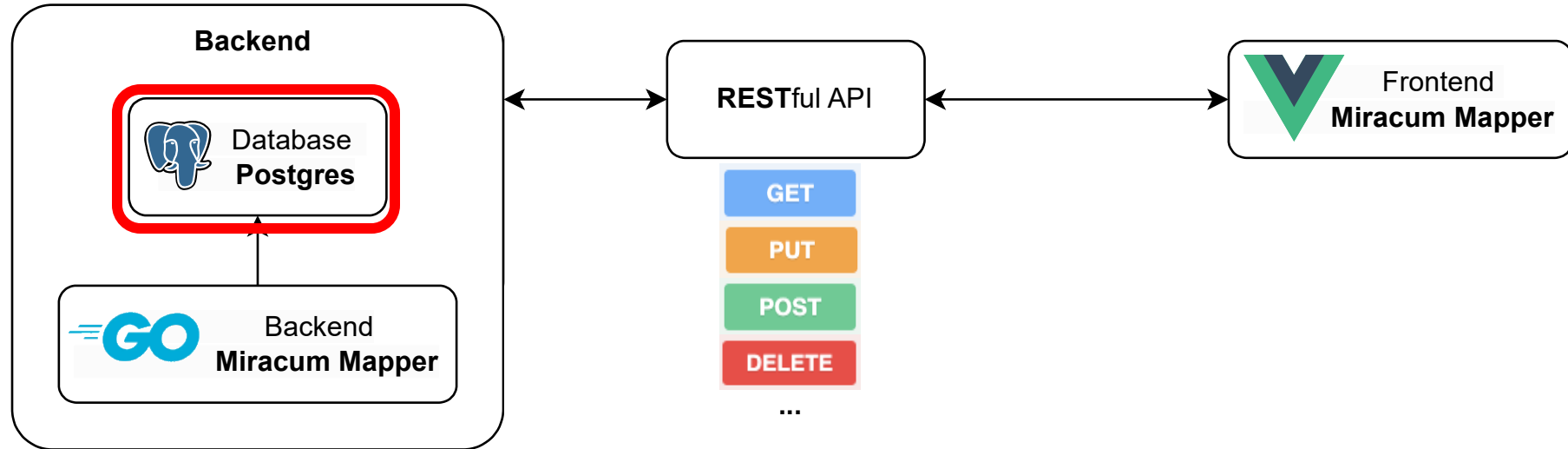
The screenshot shows a web interface titled "Code System Roles". At the top left is a green button with a plus icon and the text "+ Add Role". Below this is a table with three columns: "CodeSystem", "Role", and "Name". There are two rows of data. The first row has a menu icon, a dropdown for "Loinc 1.0", a dropdown for "Source", a text input field containing "Lab Codes", and a red trash icon. The second row has a menu icon, a dropdown for "Snomed 1.0", a dropdown for "Target", a text input field containing "Name", and a red trash icon.

Codesystem-Auswahl bei der Projekterstellung

Miracum Mapper 2.0

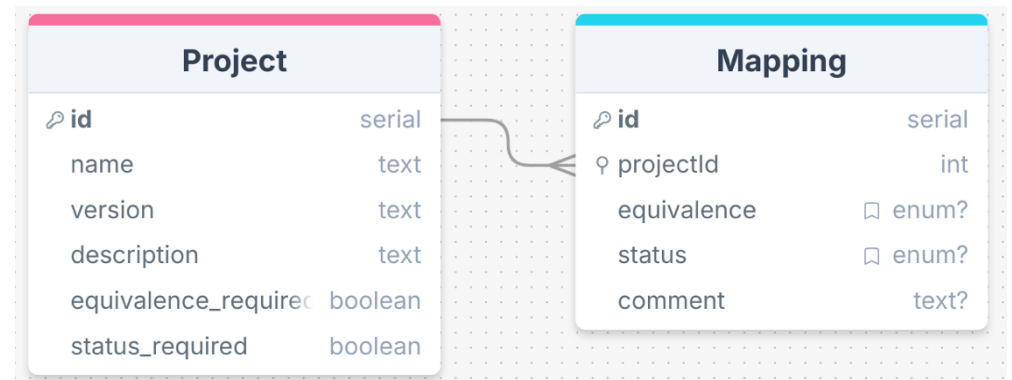


- Webanwendung: Backend und Frontend, Kommunikation mittels API
 - Keine Installation notwendig
 - Zentraler Speicherort für alle Mapping-Daten
 - Möglichkeit, an verschiedenen Geräten synchronisiert zu arbeiten

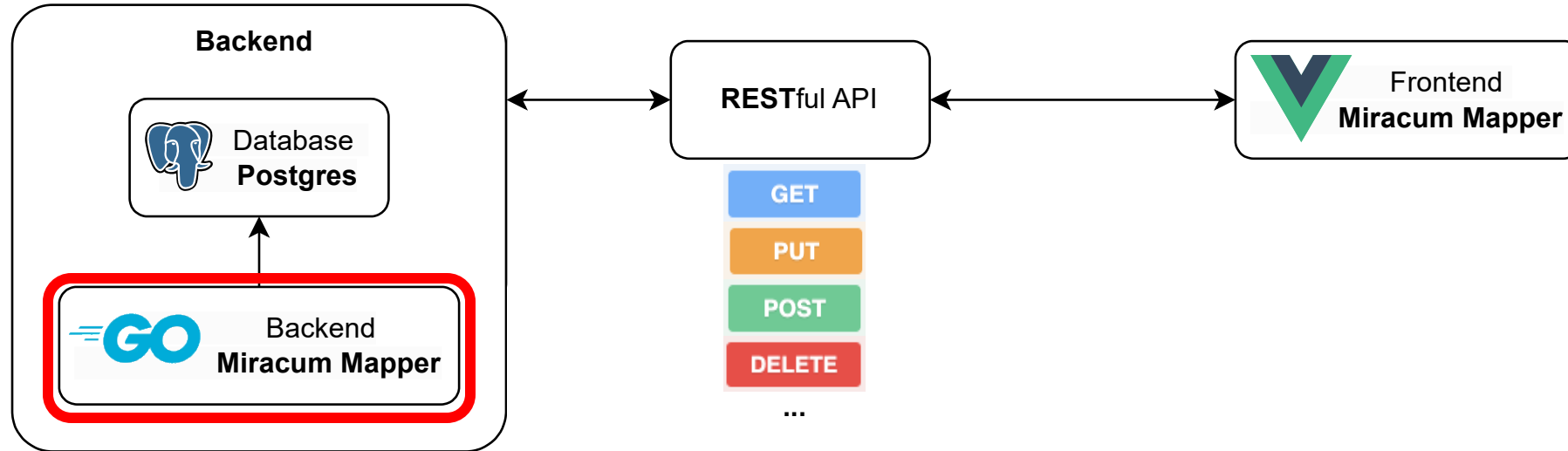


- Relationale Datenbank: Postgres

→ Schnelle Volltext- oder Ähnlichkeitssuche

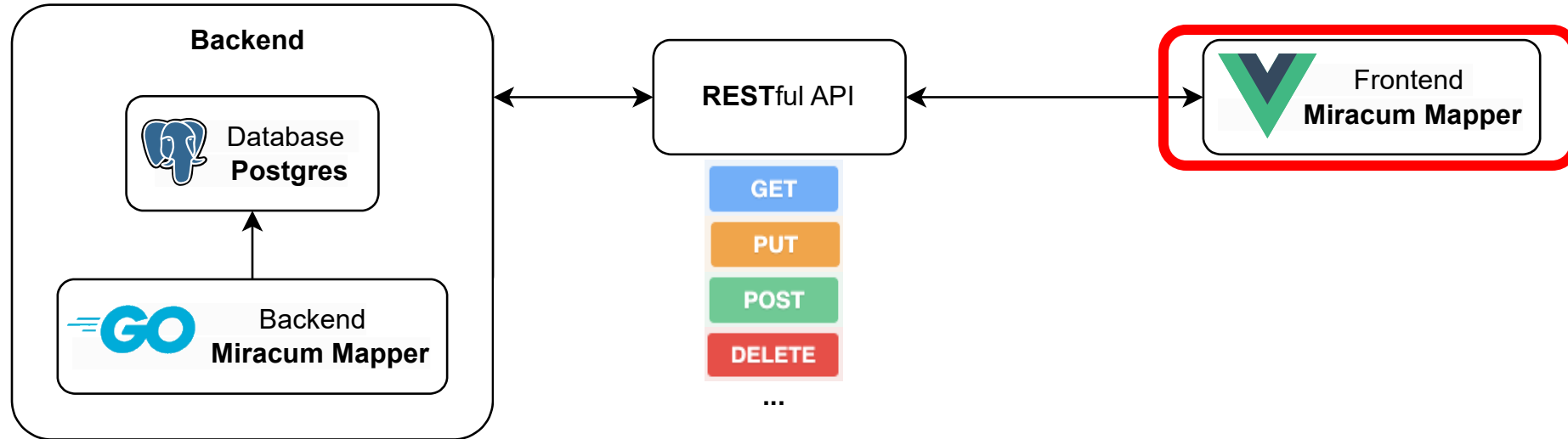


Ausschnitt aus Datenbankschema



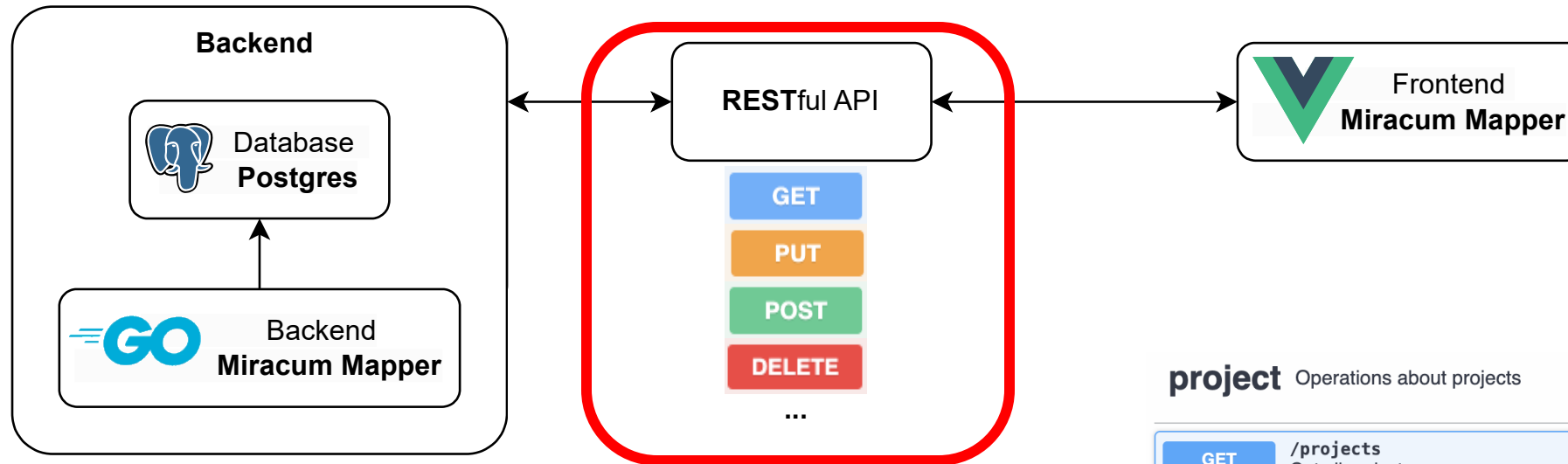
- Backend in Go → Kompilierte Sprache: sehr schnell, Typsicherheit
 - Definition des Datenbankschemas, Zugriff über ein ORM
 - Serverfunktionen, „Schnittstelle“ zwischen API und Datenbank





- Frontend mit Vue.js (Webframework)
- Typescript: Typsicherheit
- Komponentenbibliothek: PrimeVue (vorgefertigte UI-Komponenten)
- Zwischenspeicherung von Daten: Pinia
- ...

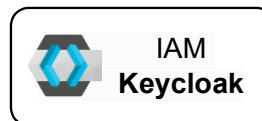




- Kommunikation mittels API
 - Definition von Endpunkten mittels OpenAPI Spezifikation in YAML
 - Visualisierung / Testen durch Swagger UI



- Außerdem: Nutzerverwaltung mit Keycloak



project Operations about projects ^		
GET	/projects Get all projects	🔒 ▼
POST	/projects Create a new project	🔒 ▼
PUT	/projects Update the project information	🔒 ▼
GET	/projects/{project_id} Get project details	🔒 ▼
DELETE	/projects/{project_id} Delete a project	🔒 ▼

Beispielhafte Endpunkte

POST **/codesystems/{codesystem_id}/import** Import concepts for a code system by ID  

Import concepts for a code system by ID. The concepts are imported from a CSV file. The format of the CSV must be "code,meaning"

Parameters Try it out

Name	Description
codesystem_id * required <code>integer(\$int32)</code> <i>(path)</i>	The ID of the Codesystem codesystem_id

Request body required text/csv 

Example Value | **Schema**

```
code,meaning
123,test meaning 1
456,another meaning
789,third meaning
```

Import eines Codesystems über die API

Project Overview

Sort by Name ▾

≡

⊞

+ Add Project

Labor Codesystem to LOINC

1.0.1

Mapping the Labor Codesystem to LOINC

ICD-10-GM to SNOMED CT

2.0.0

A project to map ICD-10 codes to SNOMED CT

1:2 Mapping Project

1.0.0

A project with one source and two targets

10 ▾

<< < 11 to 13 of 13 > >>

Projekt-Übersicht

Mapping Table

Edit project

Export

+ Add

Delete

Hidden Columns

Clear

Keyword Search

	Source Snomed Source Code	Source Snomed Source Meaning	Target Loinc Target Code	Target Loinc Target Meaning	Status	Equivalence	Comment	
☐	Search by code	Concept 1	Search by code	Search by meaning	Any	4 items selected	Search by comment	
☐	1	Concept 1	1002-5	Glucose level in Blood	Active	Related To	test comment	
☐	1	Concept 1	2	blood Hemoglobin A1c/Hemoglobin.total in Blood 1000-9 Hemoglobin A1c in Blood by HPLC 1001-7 Glucose level in Blood 1002-5 Urea Nitrogen level in Blood	Active	Equivalent	please review	
☐	1	Concept 1	2			Related To	comment	
☐	1	Concept 1	1000-9			Related To		
☐	1	Concept 1	1002-5	Glucose level in Blood	Active	Equivalent	test comment	

<<

<

1

>

>>

Showing 1 to 7 of 7 mappings

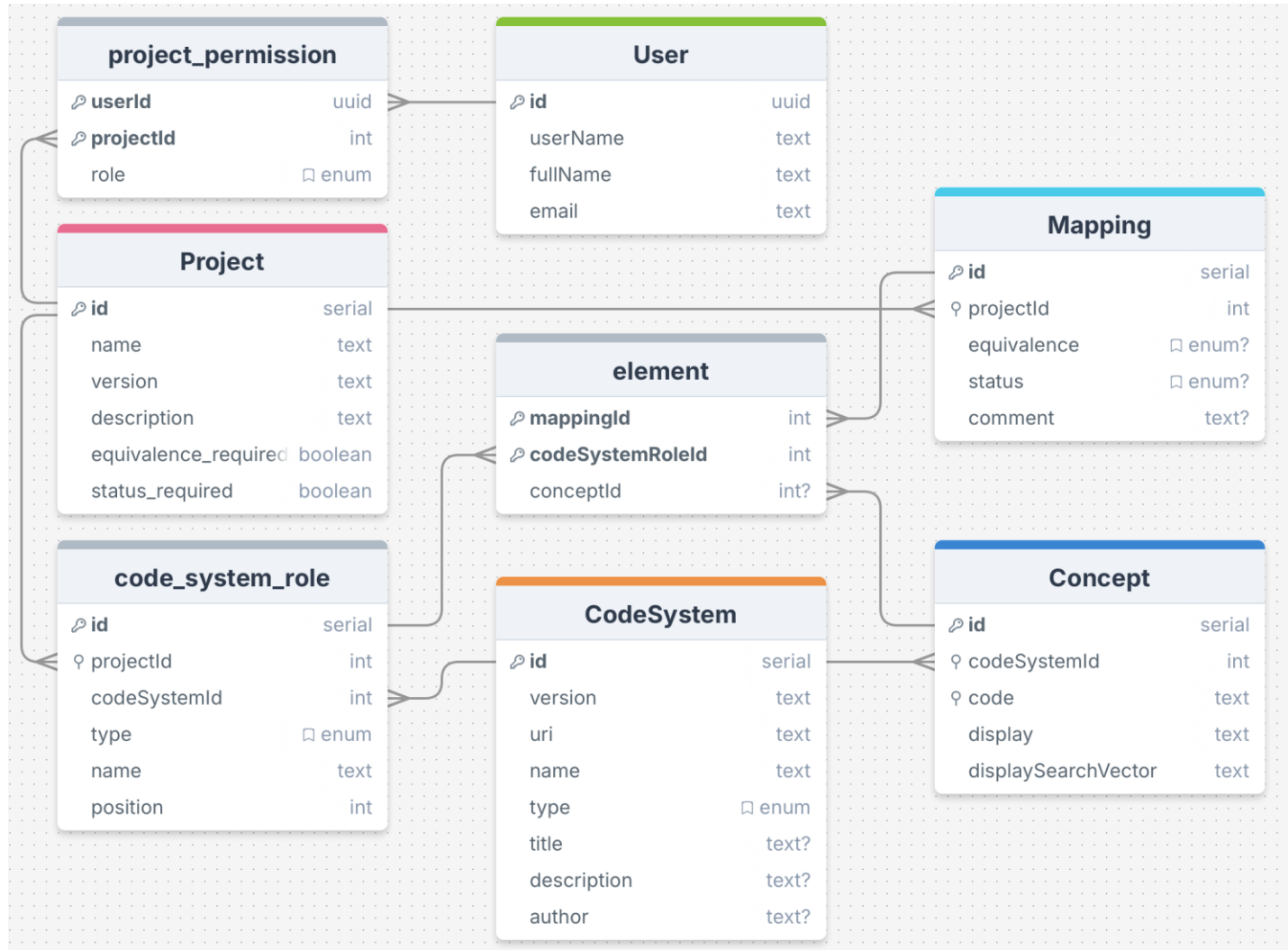
10

Miracum Mapper 2.1

Implementierung und Live Demo

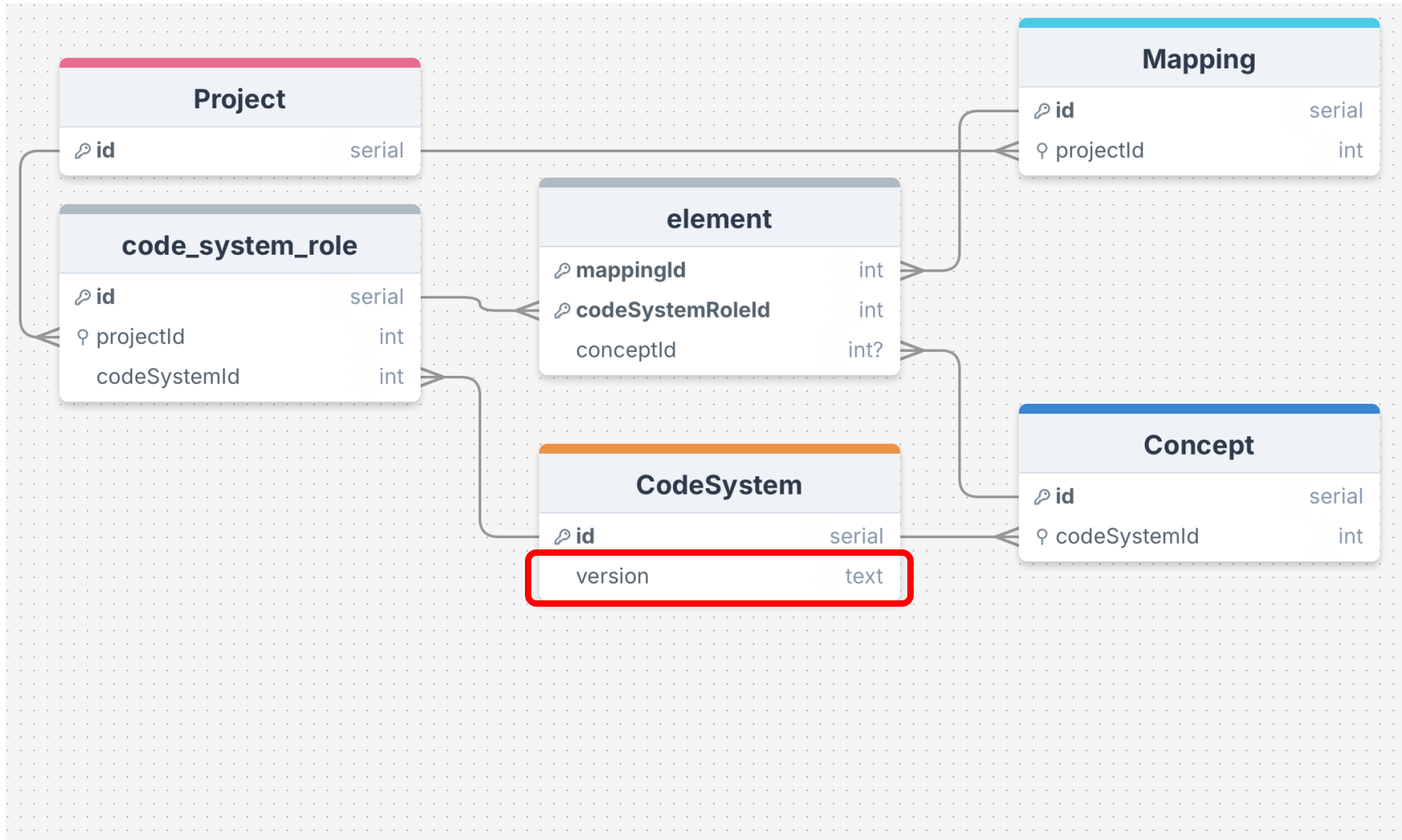
Implementierung

Anpassung des Datenbankschemas



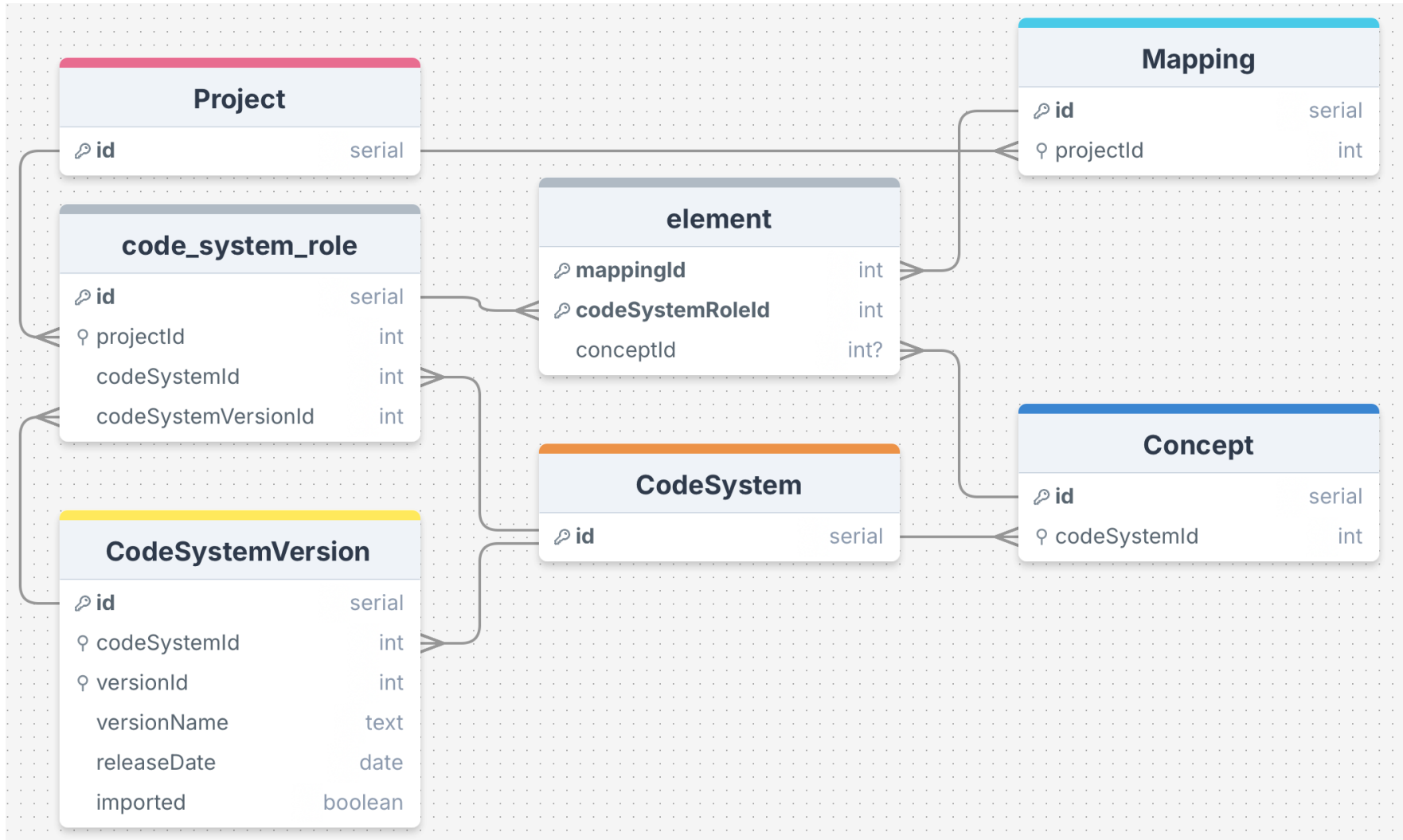
Implementierung

Anpassung des Datenbankschemas



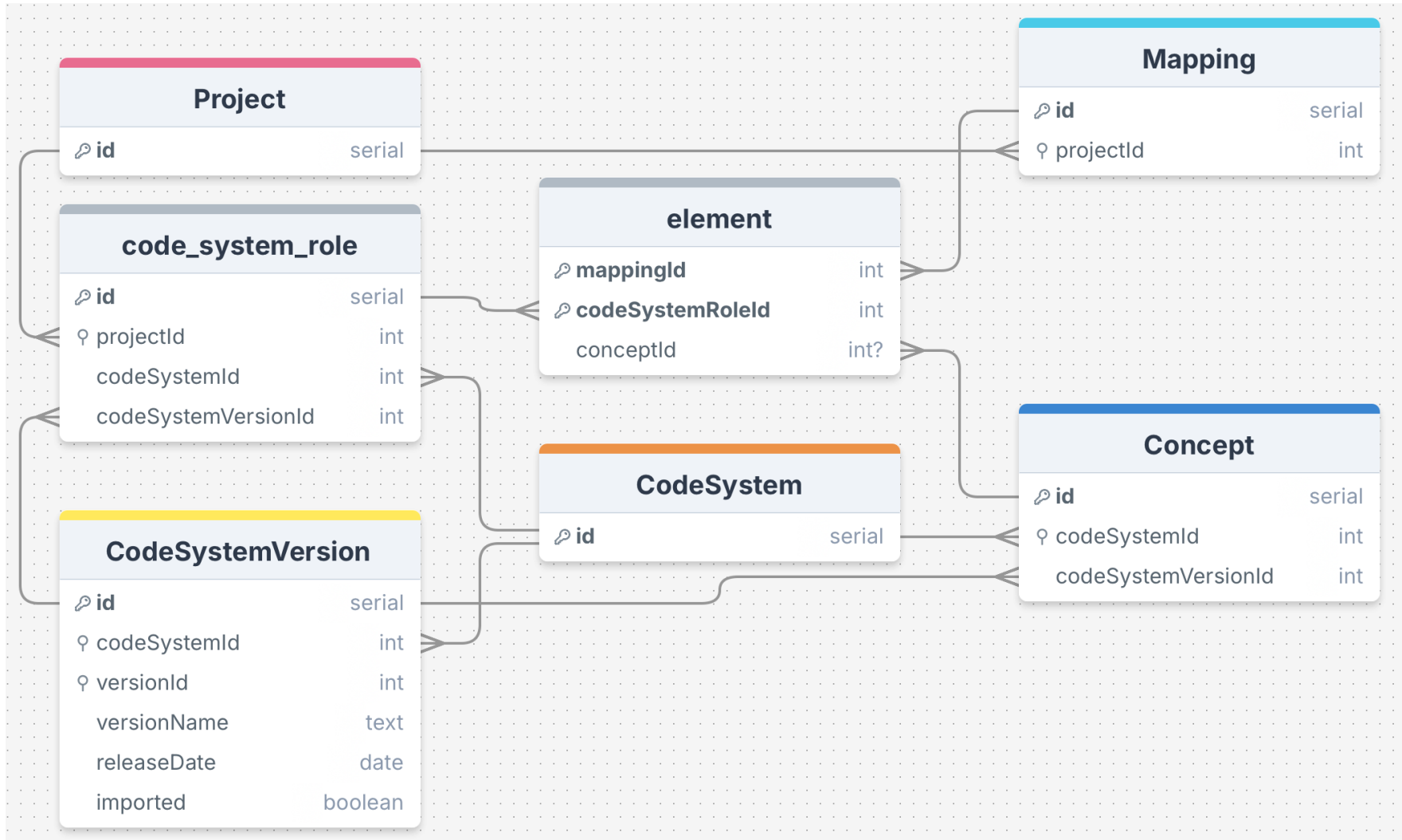
Implementierung

Anpassung des Datenbankschemas



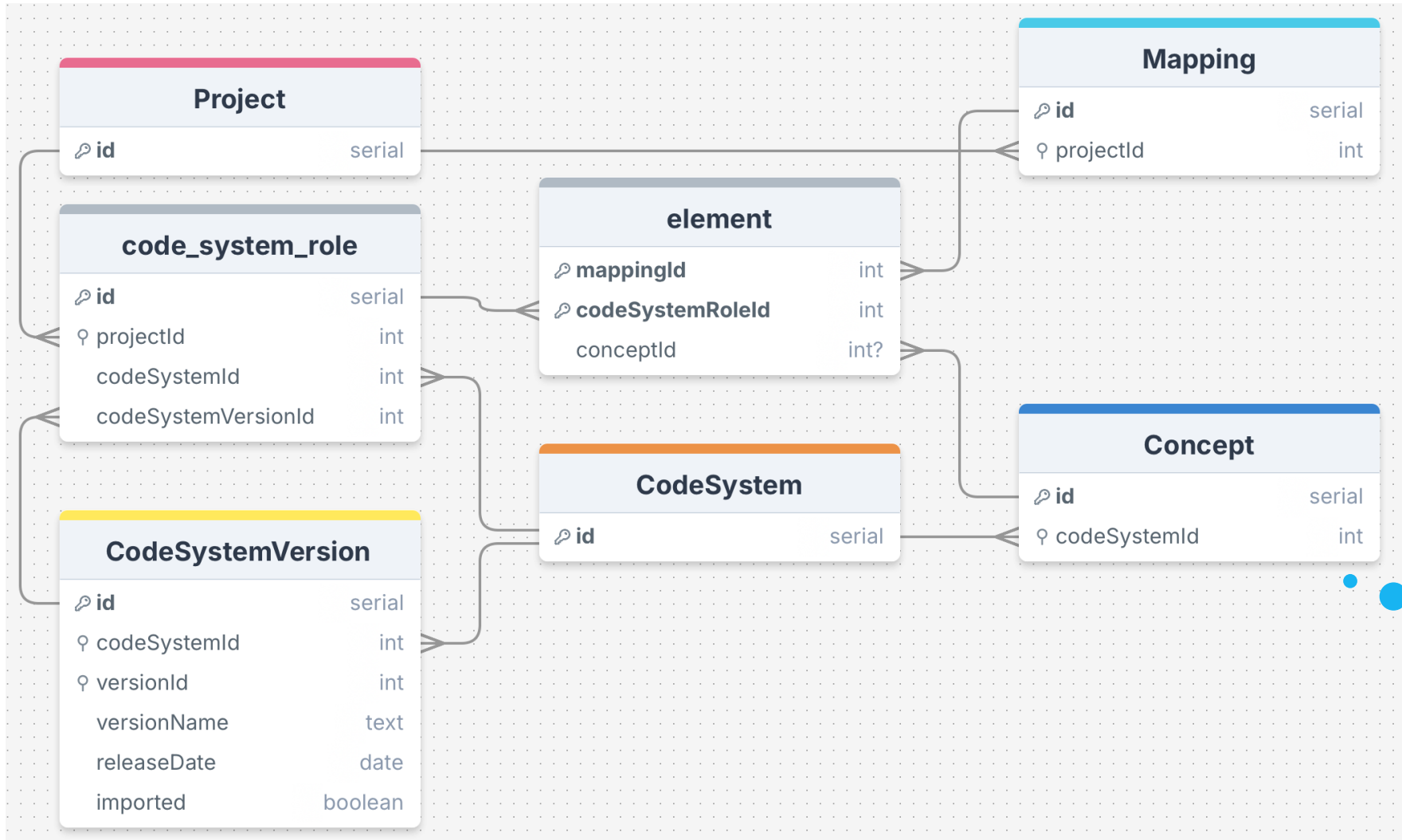
Implementierung

Anpassung des Datenbankschemas

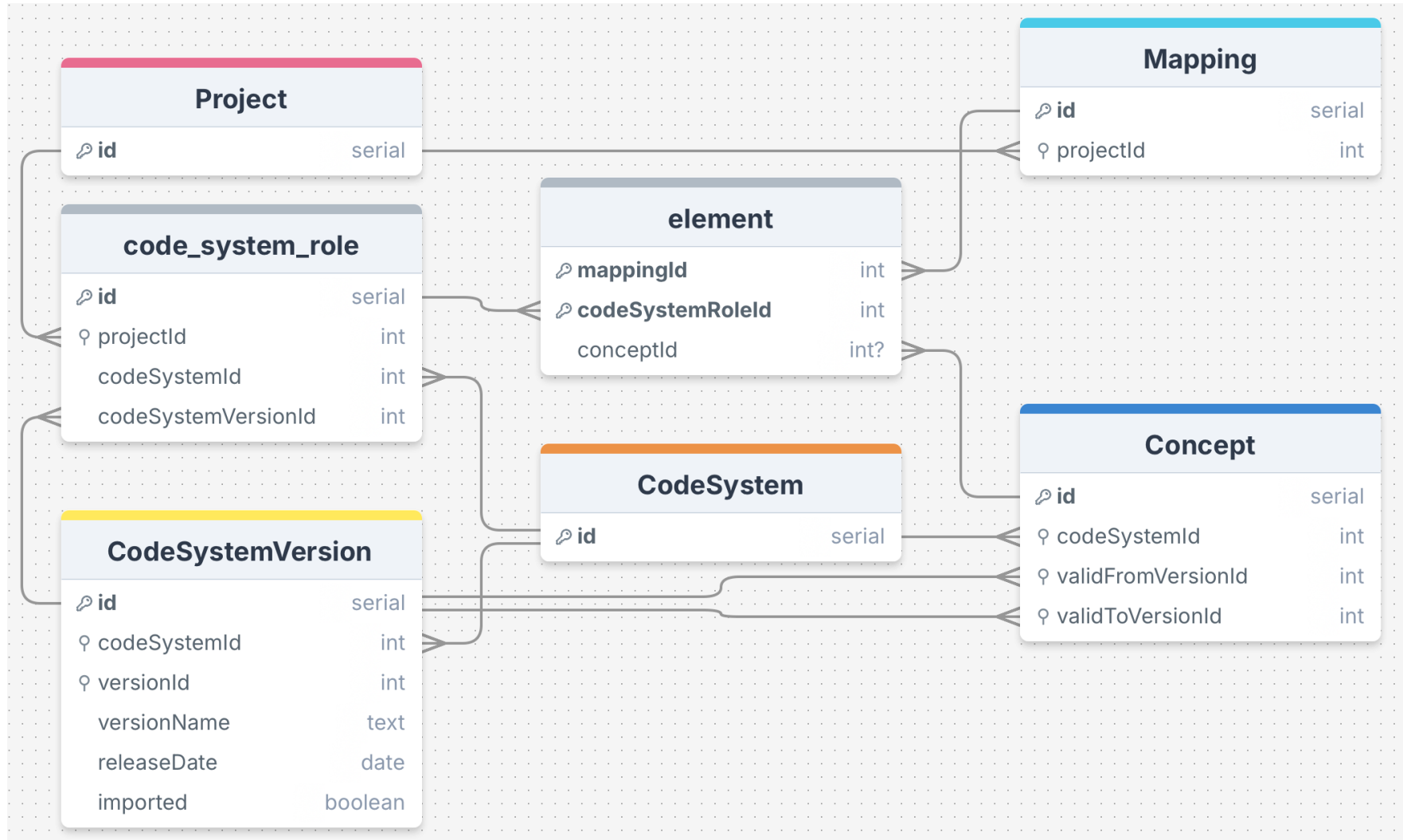


Implementierung

Anpassung des Datenbankschemas



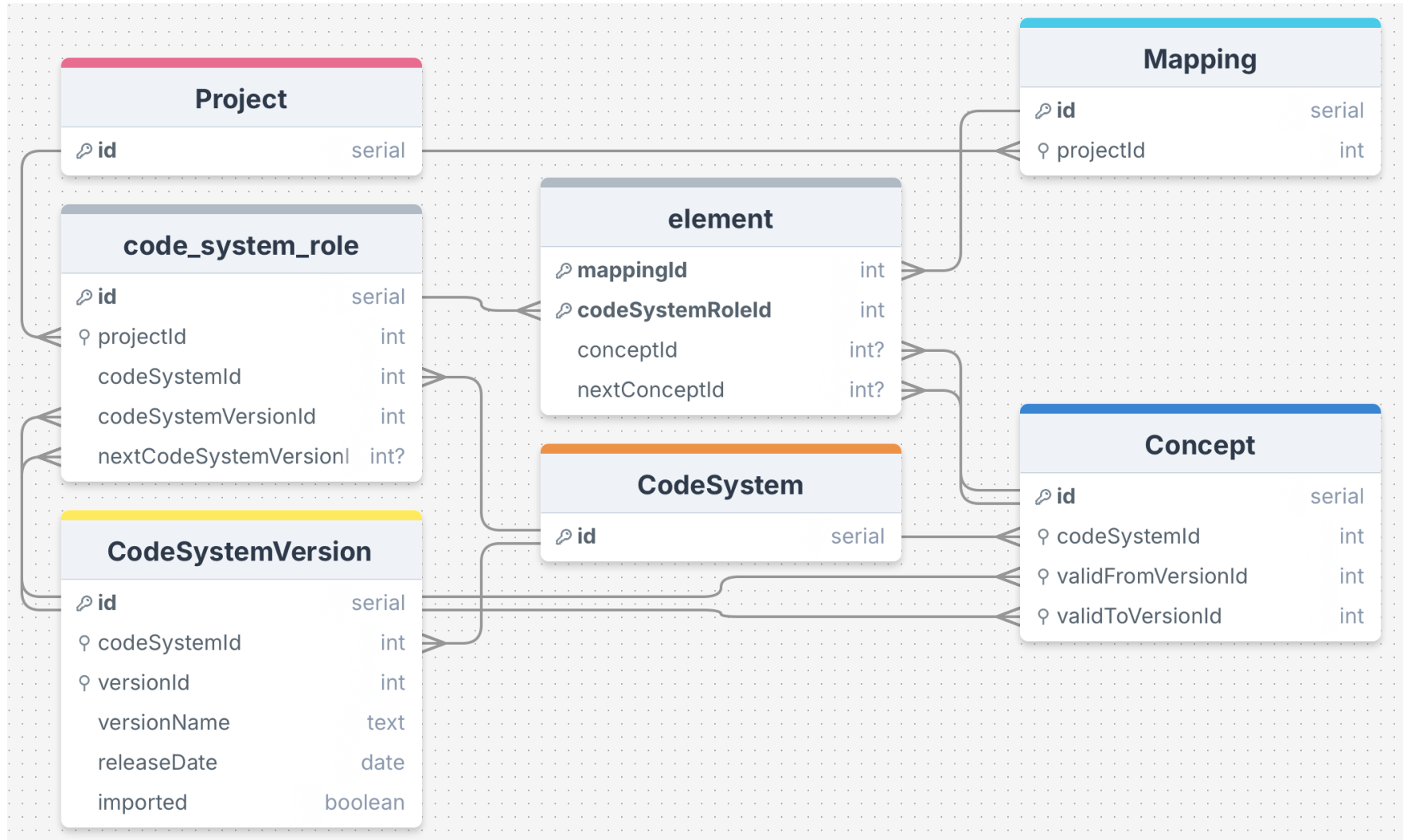
Liste von
Änderungen ?



- SNOMED CT: pro Jahr an bis zu 10% der Konzepte Änderungen
(Lee et al.: *Implications of SNOMED CT versioning*
doi:10.1016/j.ijmedinf.2011.02.006)
- LOINC: pro Jahr an bis zu 5% der Konzepte Änderungen

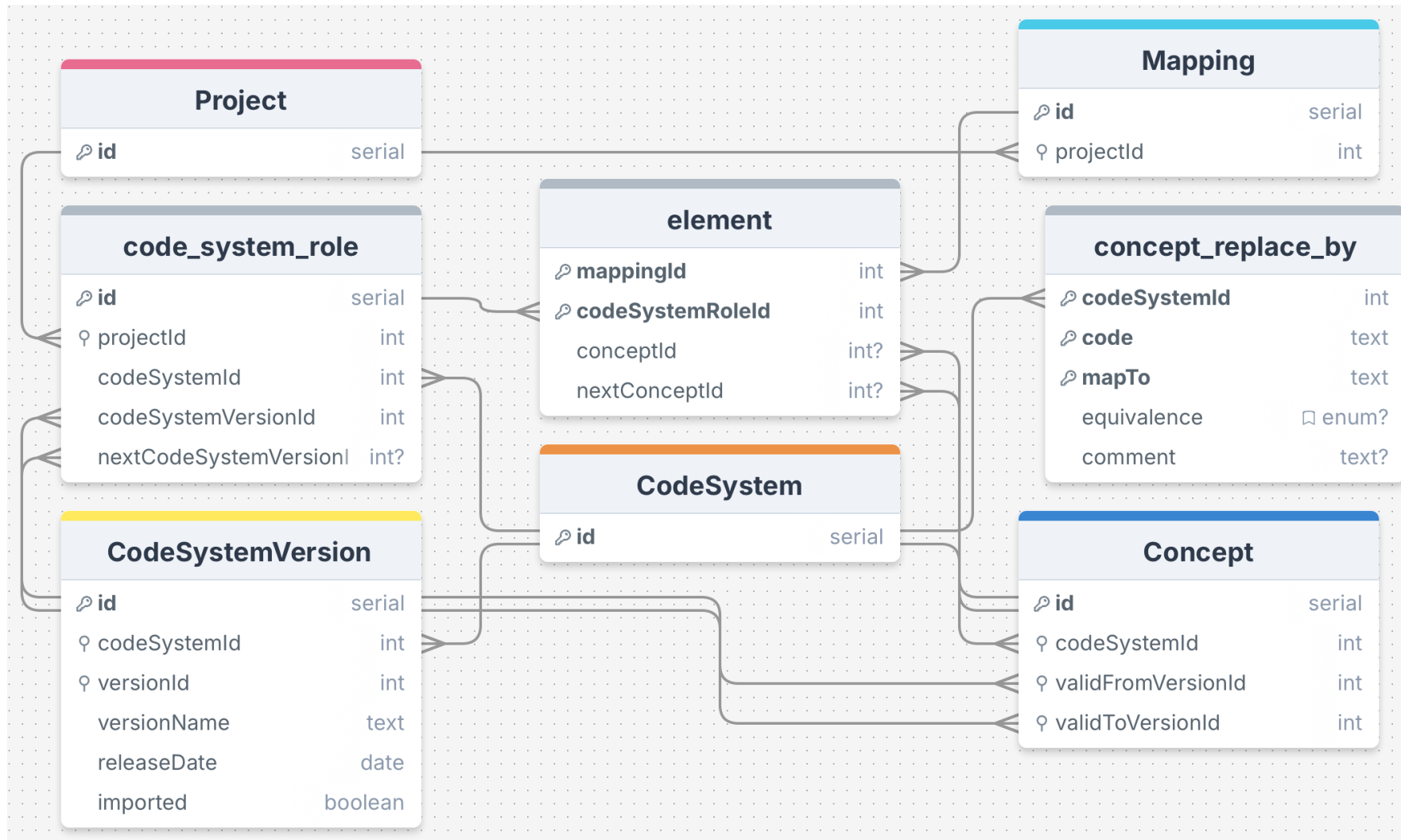
Implementierung

Anpassung des Datenbankschemas



Implementierung

Anpassung des Datenbankschemas

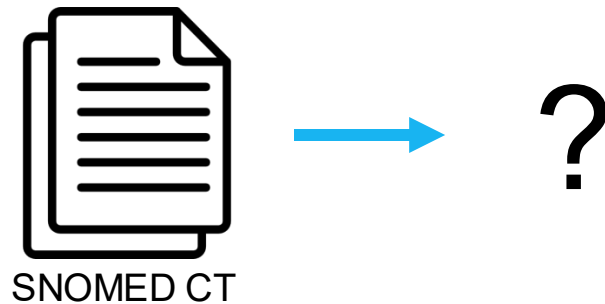
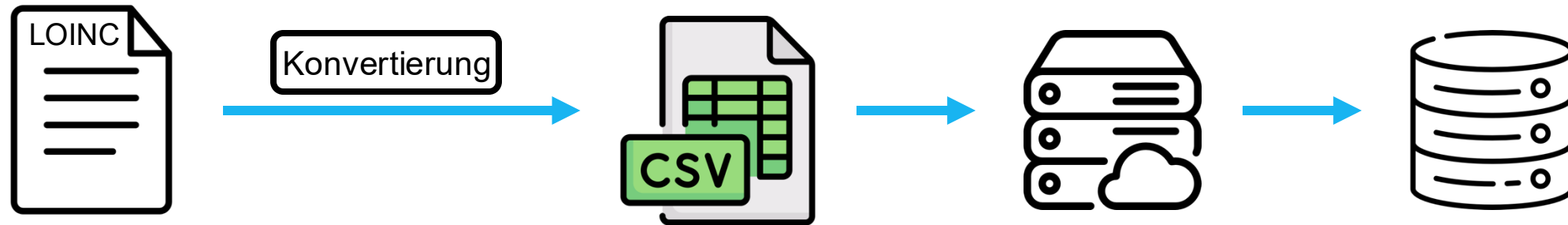


Implementierung

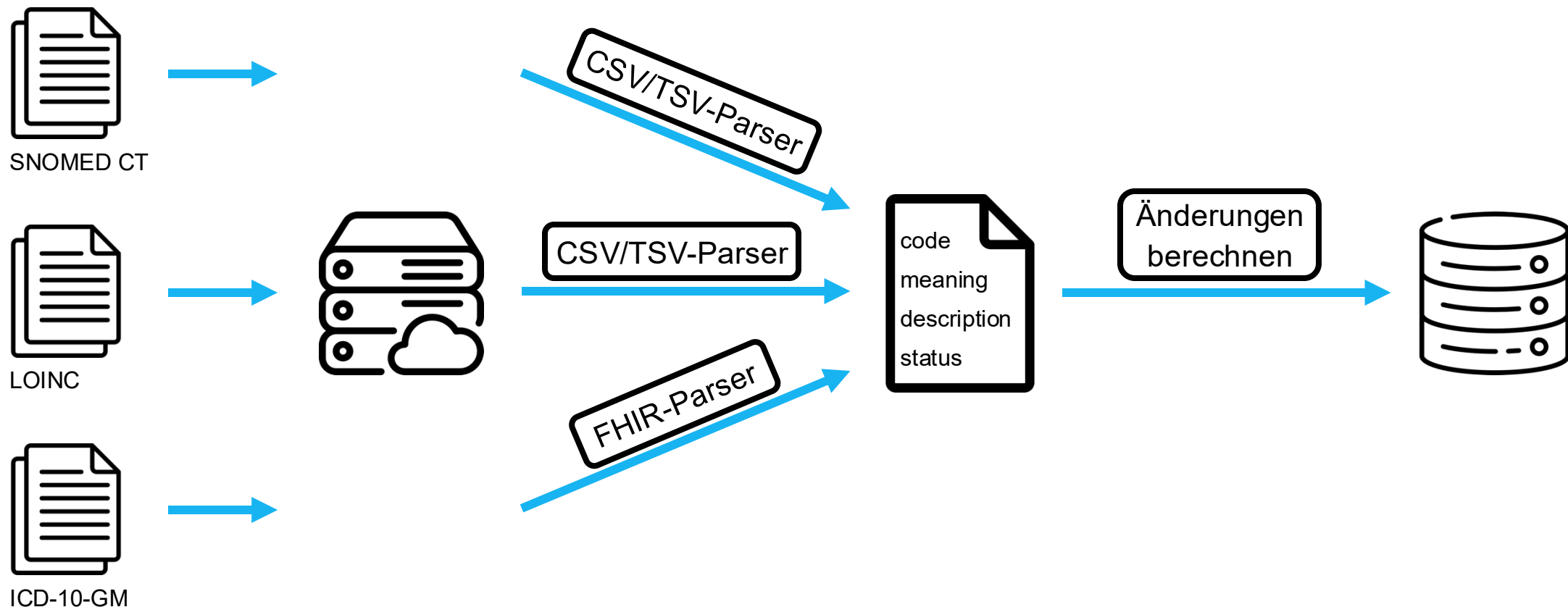
Anpassung des Datenbankschemas



- Bisher: Konvertierung in CSV-Datei mit vorgegebener Struktur notwendig

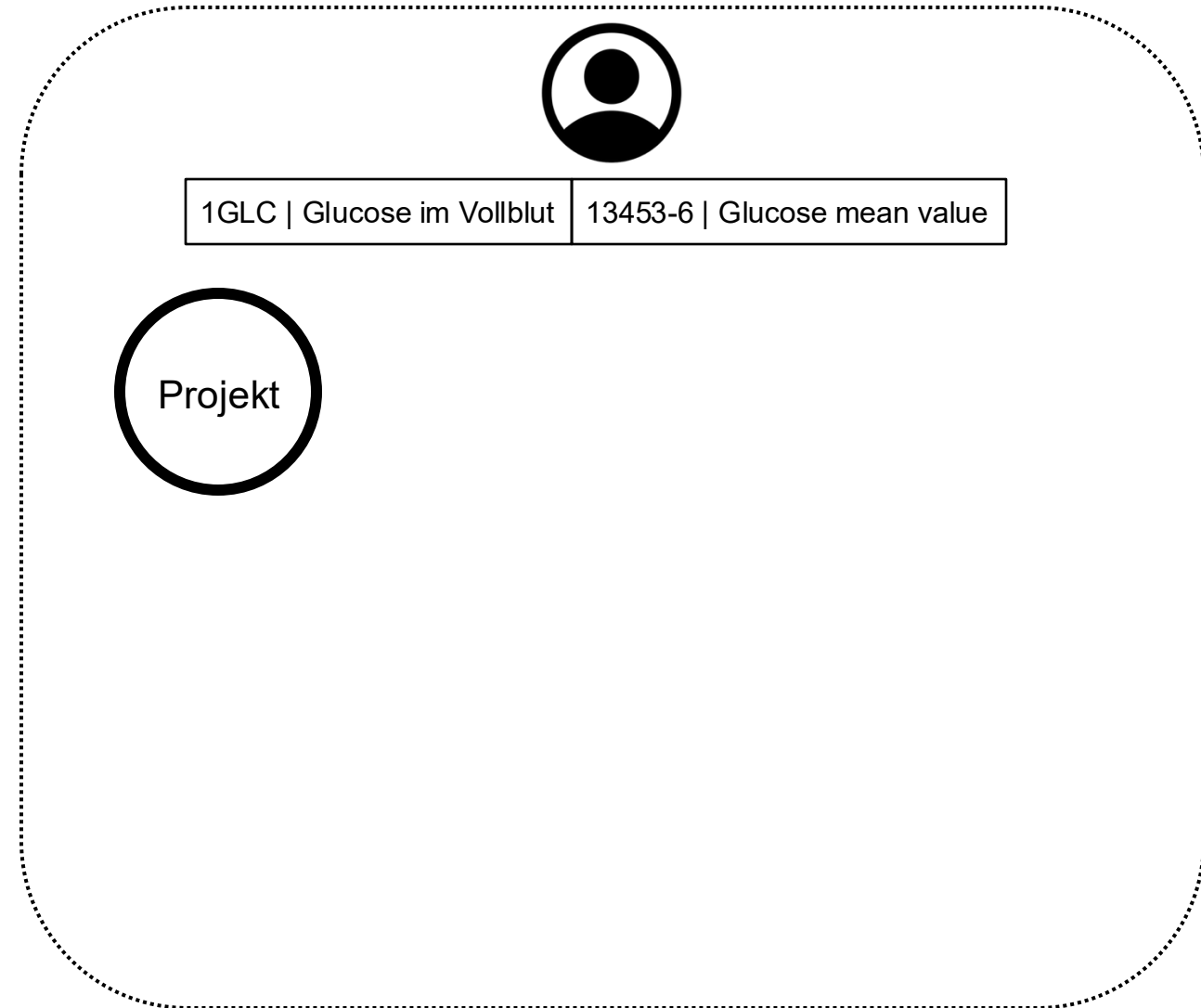


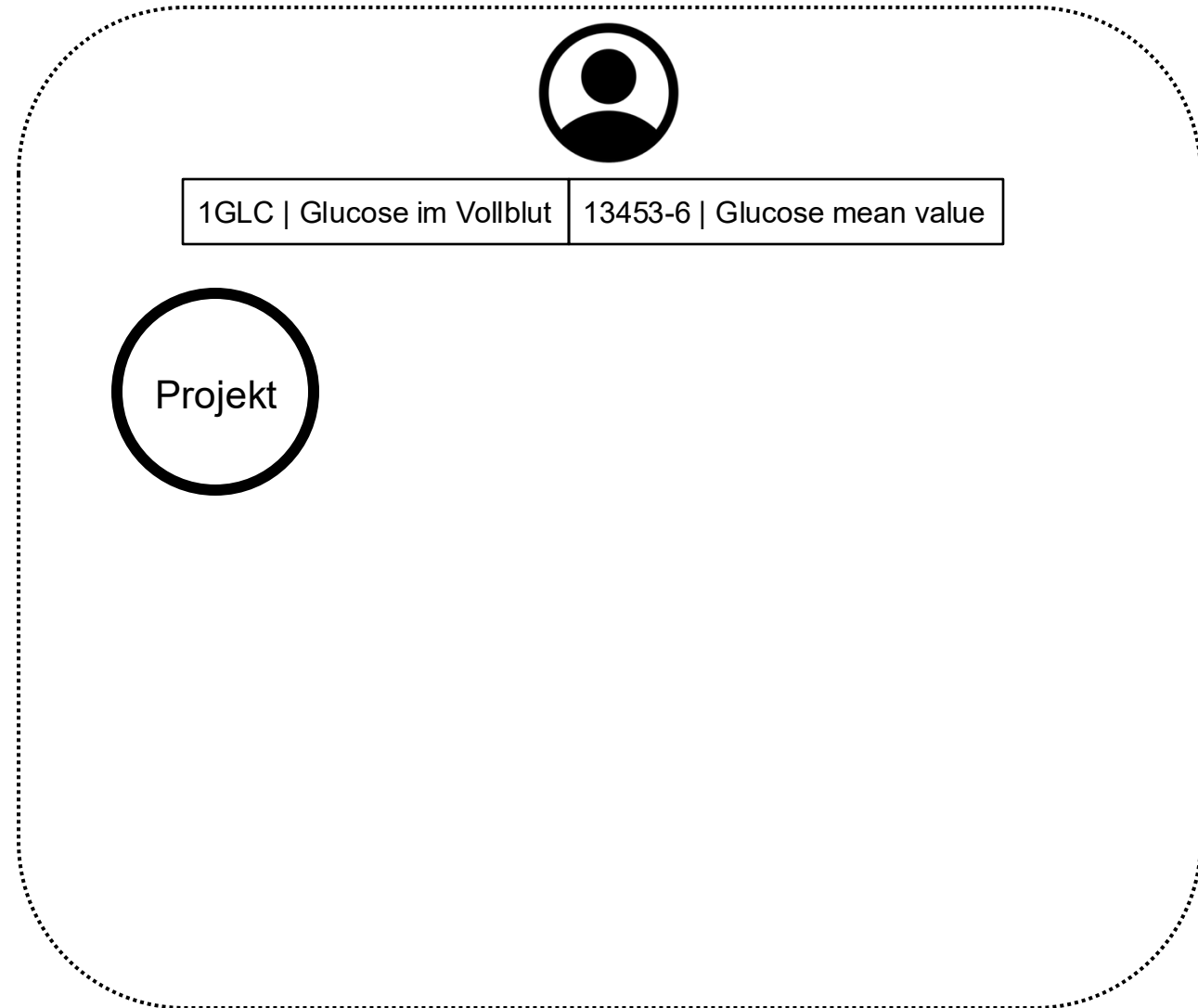
- Jetzt: Konvertierung von beliebigen Terminologien im Backend (LOINC, SNOMED CT, ICD-10-GM)



- CSV-Endpunkt immer noch verfügbar (für andere Terminologien oder Labor-Codesystem)
- Berechnung der Änderungen zwischen zwei Versionen durch Backend
 - Import dauert länger (bei SNOMED CT einige Minuten)
- Versionen müssen nicht chronologisch importiert werden

Live Demo

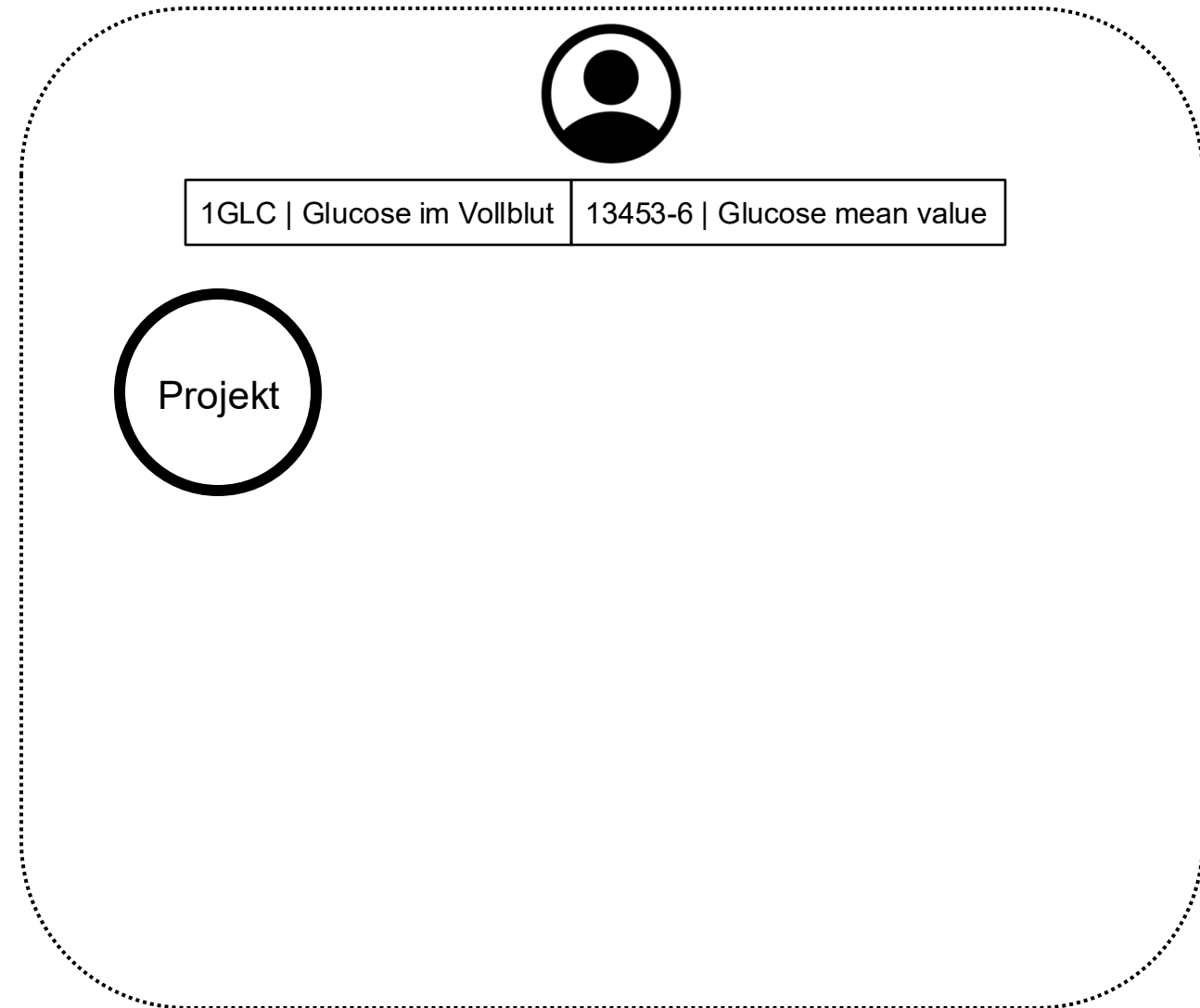






LOINC hat neuere Versionen:

- 2.78 – 06.08.2024 – ID: 2
- 2.79 – 18.02.2025 – ID: 3



Implementierung

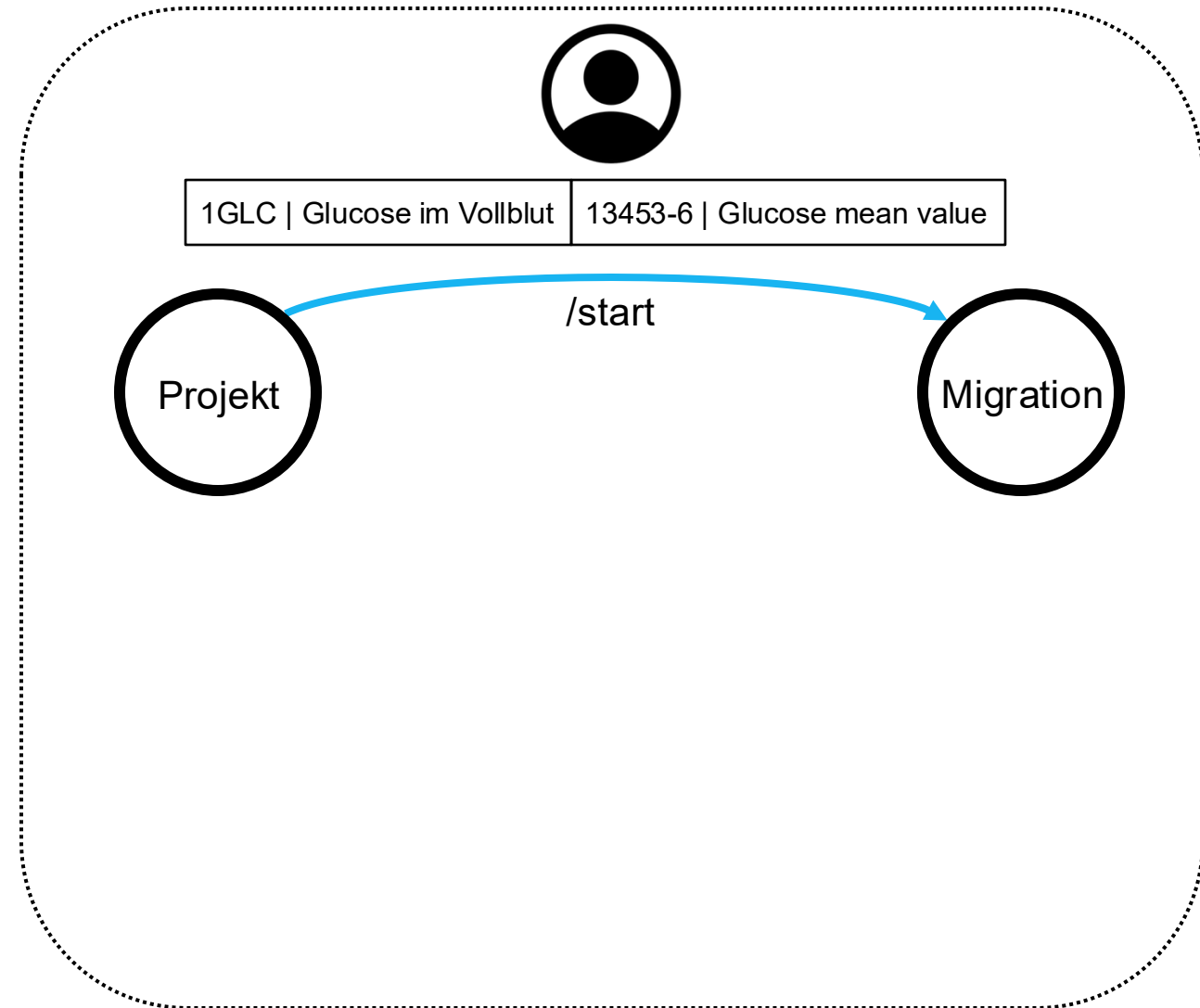
Migration eines Projekts



POST

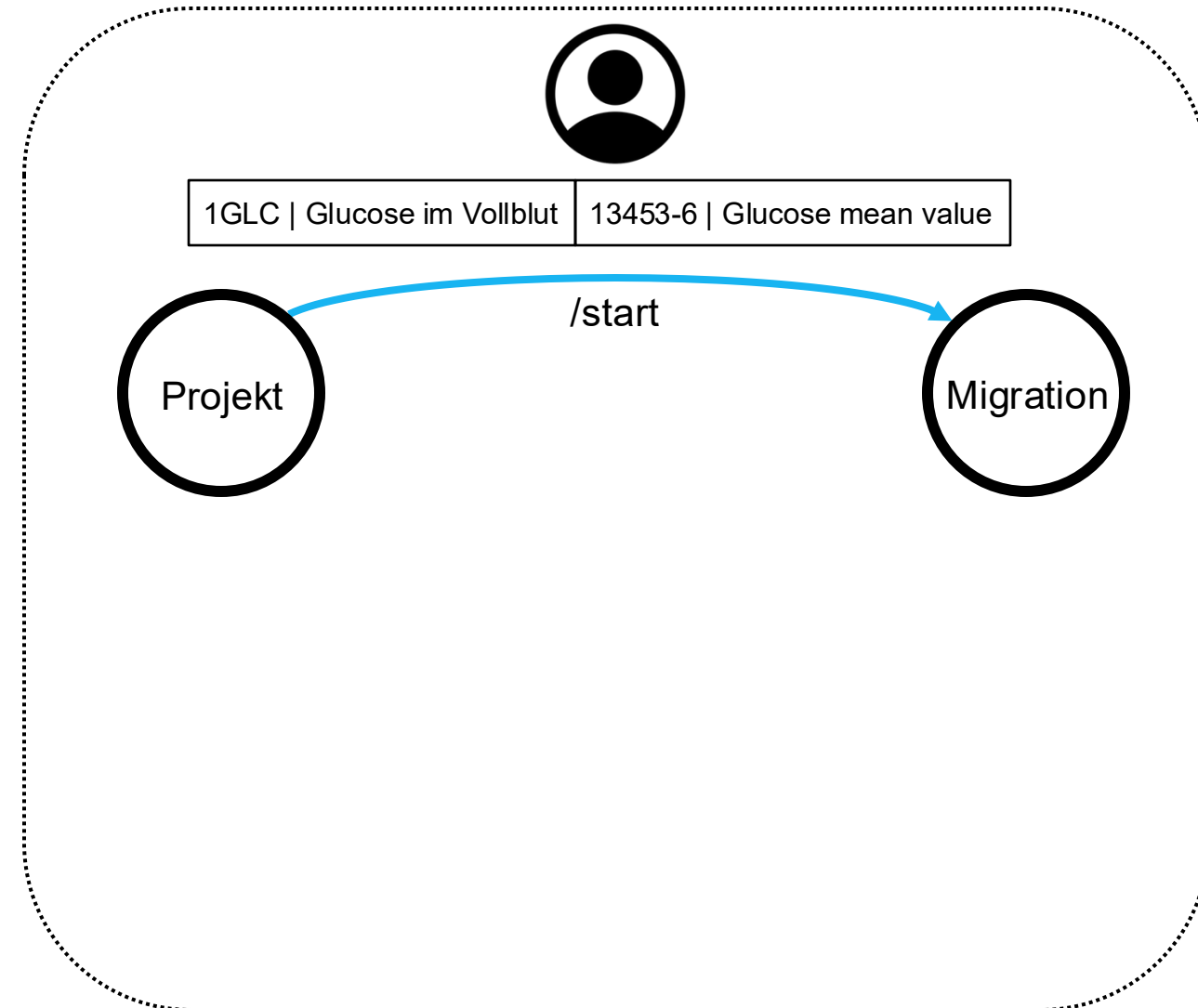
/start

Migration zu ID: 3



Implementierung

Migration eines Projekts





Änderung in Mapping Nr. 42:

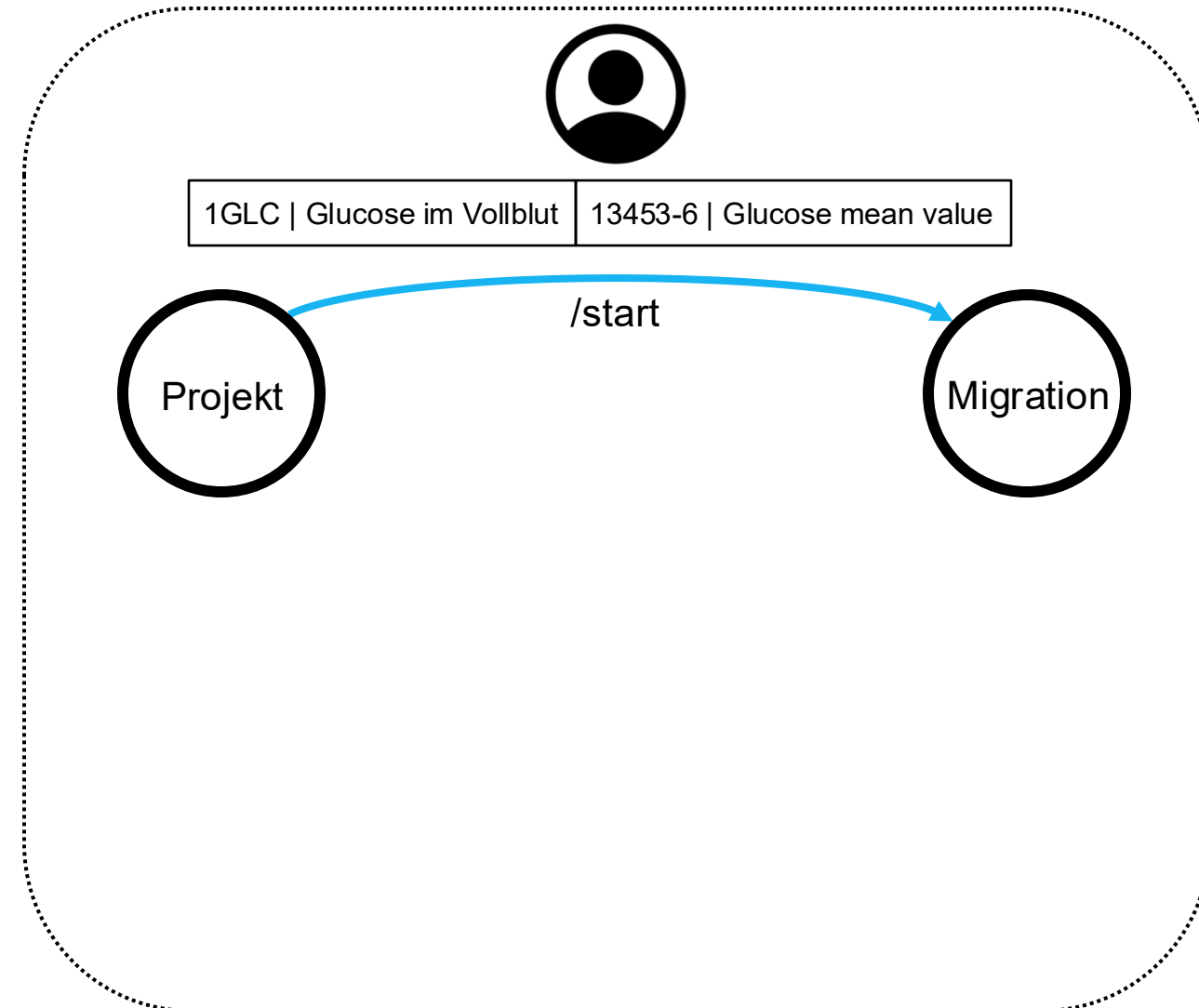
Status hat sich geändert:

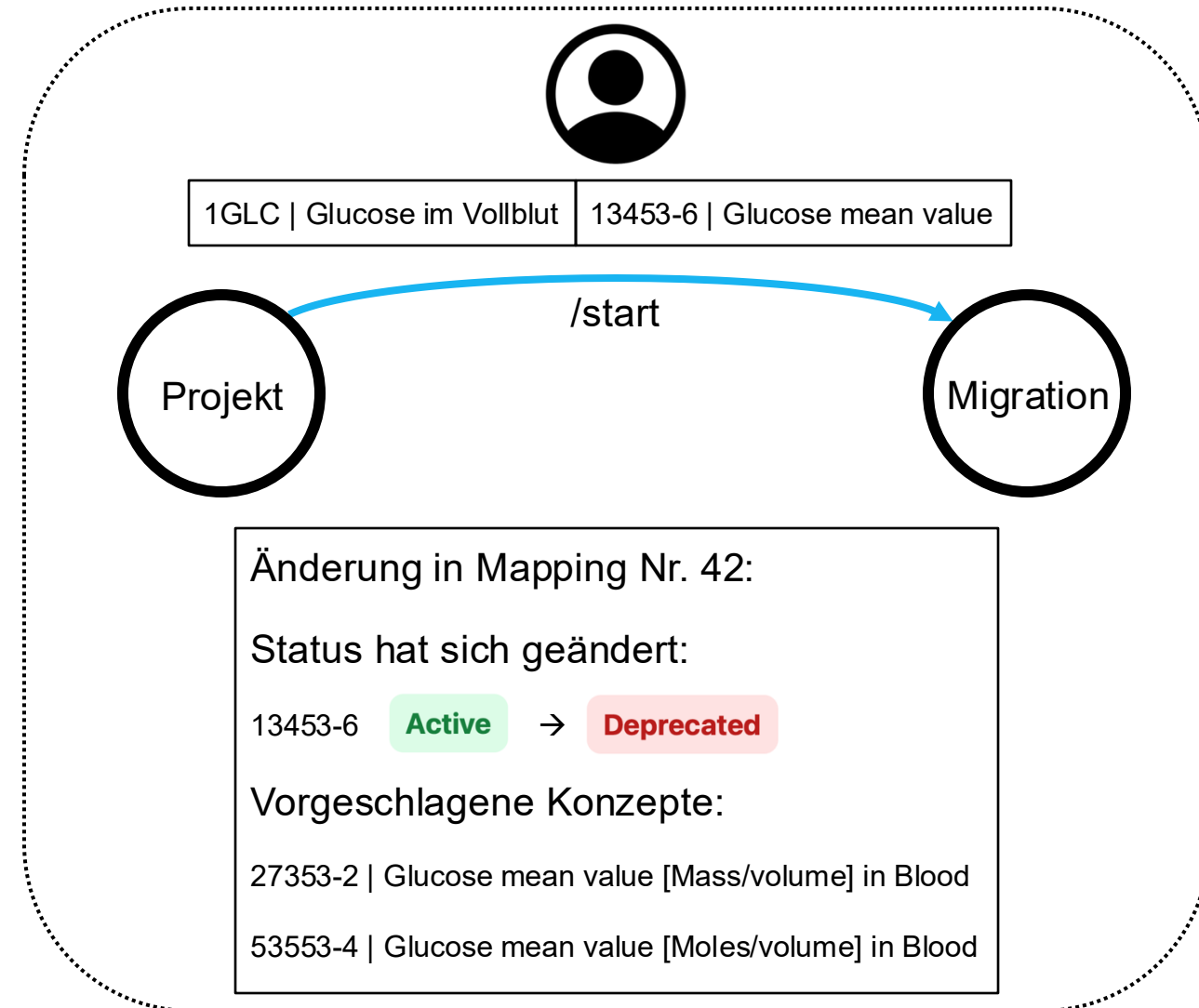
13453-6 **Active** → **Deprecated**

Vorgeschlagene Konzepte:

27353-2 | Glucose mean value [Mass/volume] in Blood

53553-4 | Glucose mean value [Moles/volume] in Blood







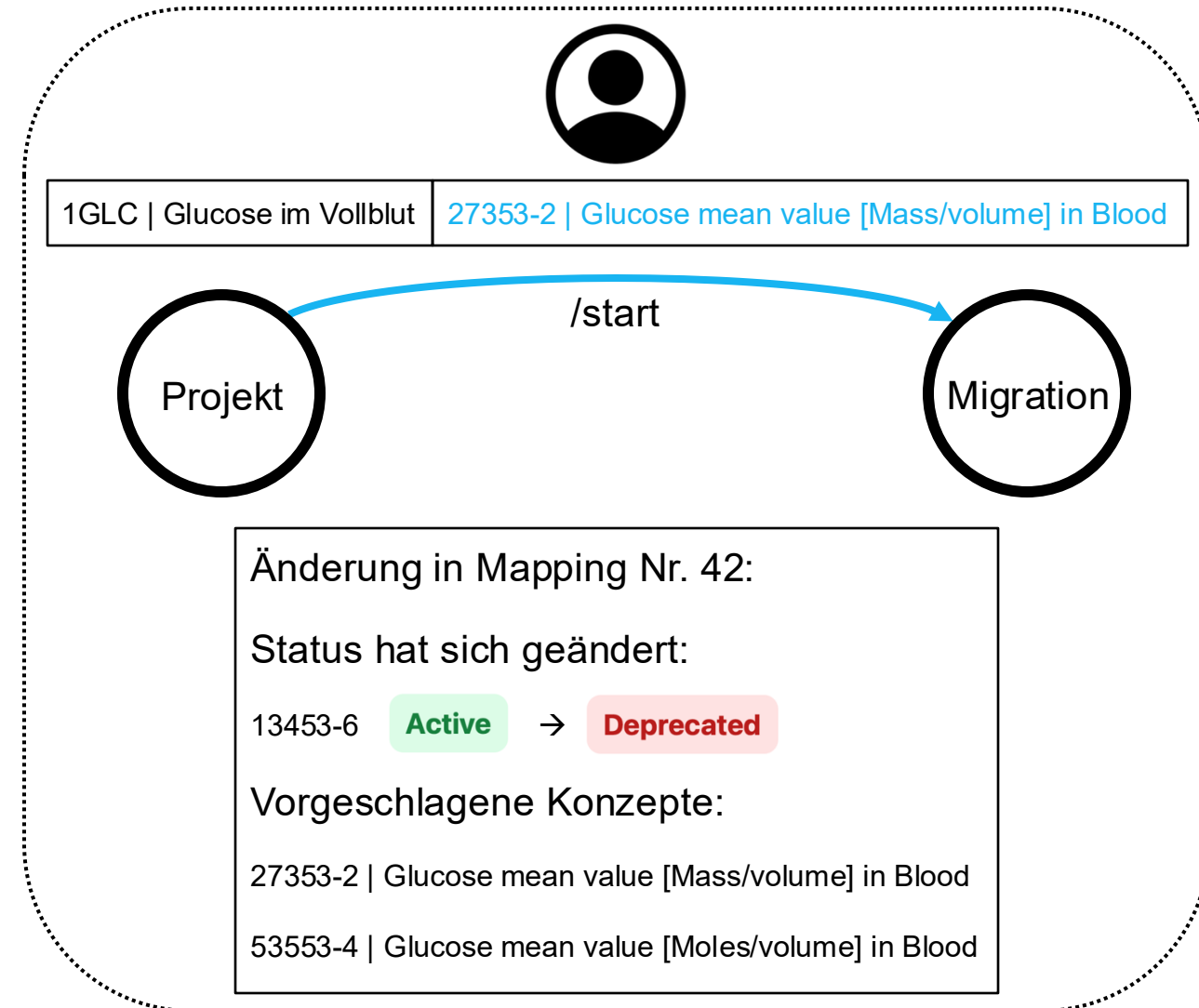
POST

/migrate

Mapping Nr. 42:

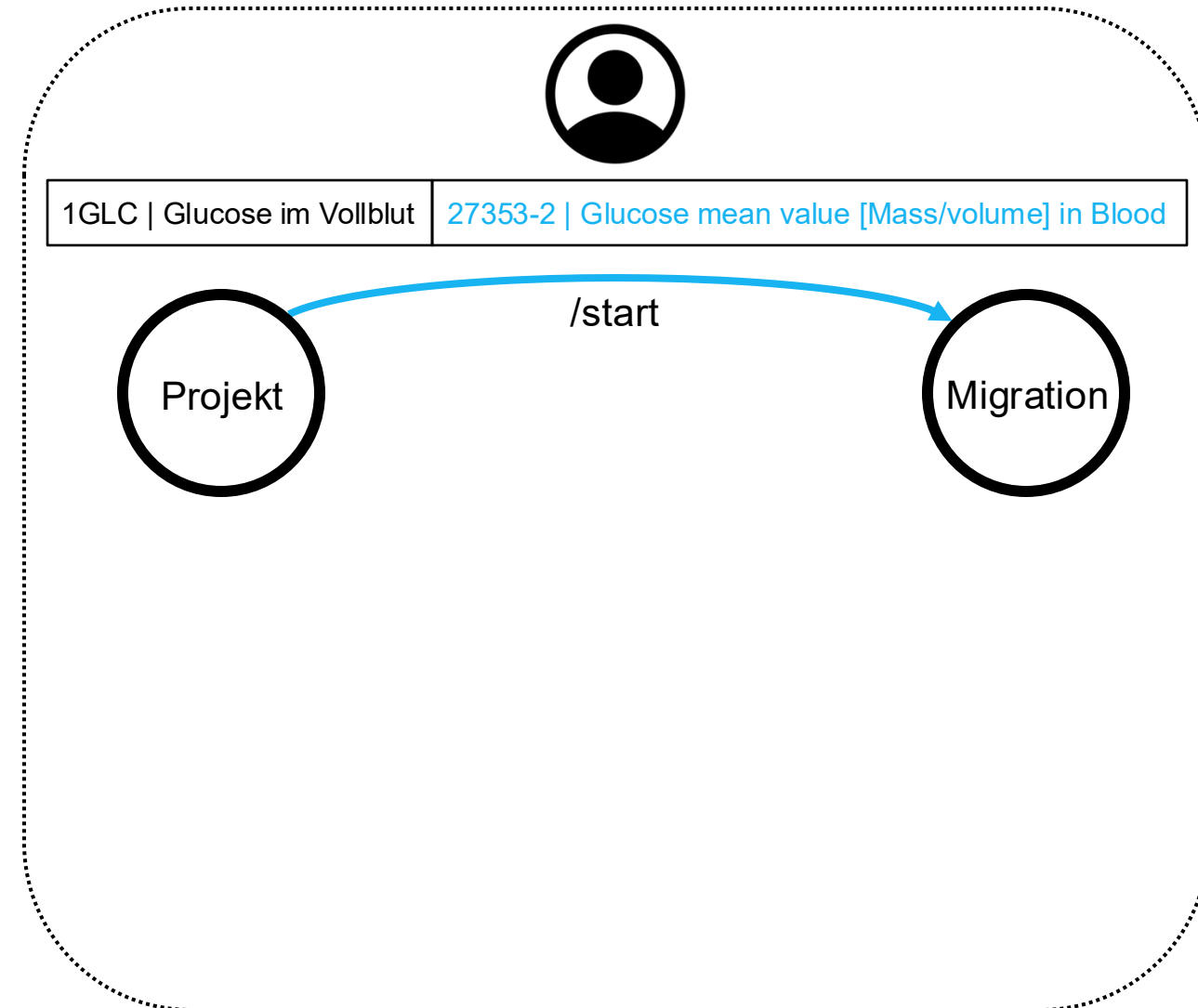
Konzept austauschen

13453-6 → 27353-2



Implementierung

Migration eines Projekts

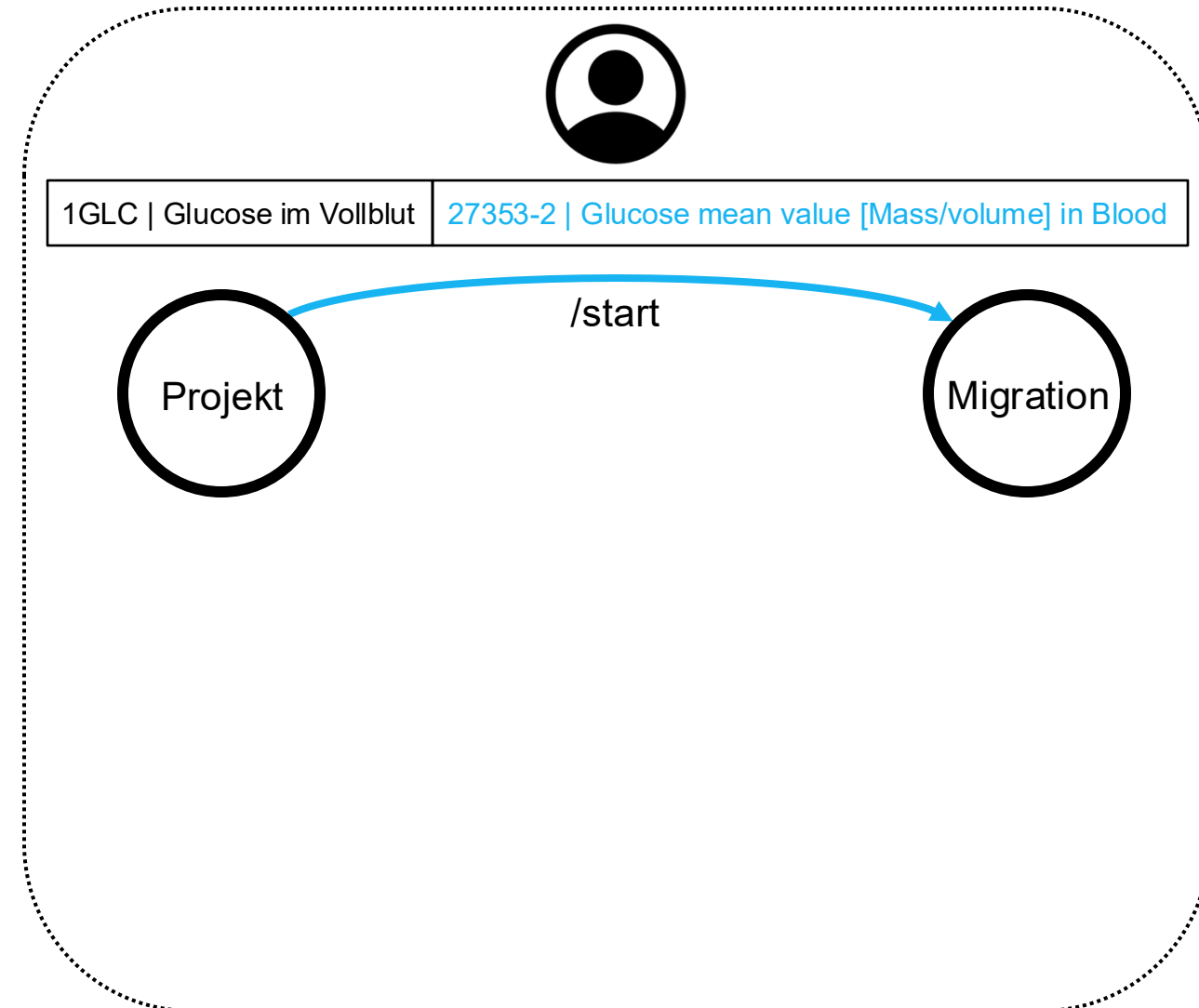


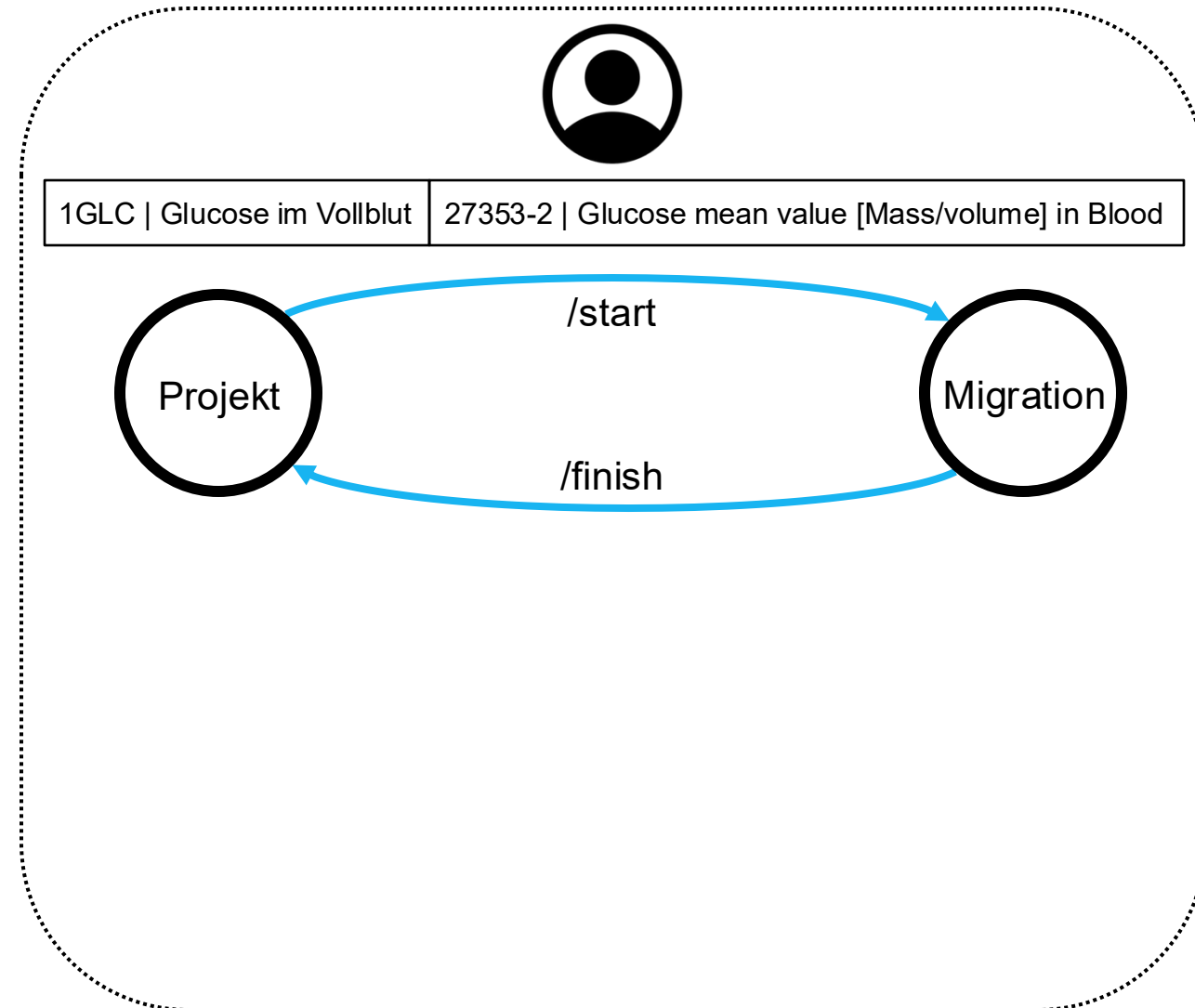
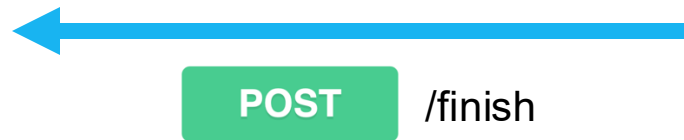
Implementierung

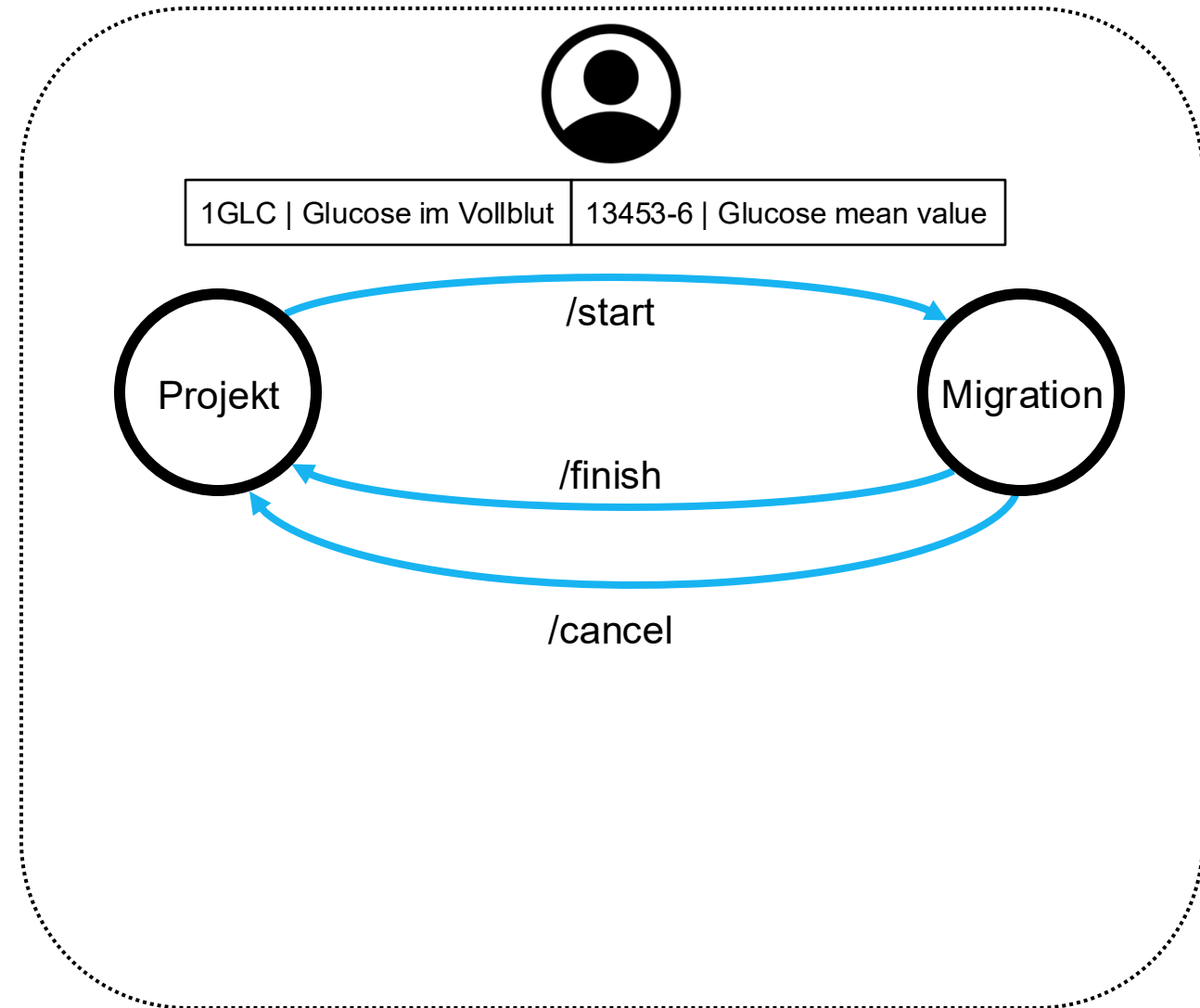
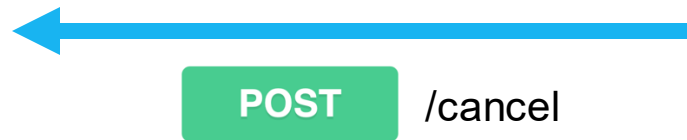
Migration eines Projekts



Keine Mappings mit Änderungen







Live Demo

Ausblick

- Miracum Mapper 2.1 enthält die wichtigsten Funktionen zum produktiven Einsatz
- Erweiterungen des Tools möglich
 - Änderungshistorie von Mappings
 - Beziehungen zwischen Konzepten (z.B. Hierarchie bei SNOMED CT)
 - Anbindung an Terminologie-Server (z.B. Benachrichtigung bei neuen Versionen, Verknüpfung mit Konzept-Browsern)
 - ...
- Technische Verbesserungen
 - Schnellerer Import einer Version
 - Automatisierte Tests
 - ...

**Vielen Dank
für Ihre Aufmerksamkeit!**

Anhang – Live Demo

 MiracumMapper

Projects Codesystems



> Codesystems

Codesystem Overview

Sort by Name ▾

☰ ☒

+ Add Codesystem

LOINC
LOINC

ICD-10-GM
ICD_10_GM

SNOMED CT International Edition
SNOMED_CT

Labor-Codesystem
GENERIC

20 ▾ << < 1 to 4 of 4 > >>

 MiracumMapper

Projects

Codesystems



Codesystems

ICD-10-GM

ICD-10-GM - Versions

Edit codesystem

+ Add

Edit

Delete

Import concepts

	Name	Release Date	Imported	In use
☐	2024	2024-01-01	No	No
☐	2025	2025-01-01	No	No

 MiracumMapper

Projects

Codesystems



Codesystems

ICD-10-GM

ICD-10-GM - Versions

Edit codesystem

+ Add

Edit

Delete

Import concepts

	Name	Release Date	Imported	In use
☐	2024	2024-01-01	No	No
☒	2025	2025-01-01	No	No

 **MiracumMapper** Projects Codesystems 

> Codesystems > ICD-10-GM > Import 2025

Import Version

Import concepts to the version **2025** of code system **ICD-10-GM**.

Code system type: **ICD_10_GM**
Version release date: **2025-01-01**

+

CodeSystem-icd10gm-<version>.json *

No file chosen 

+

ConceptMap-icd10gm-<version>.json *

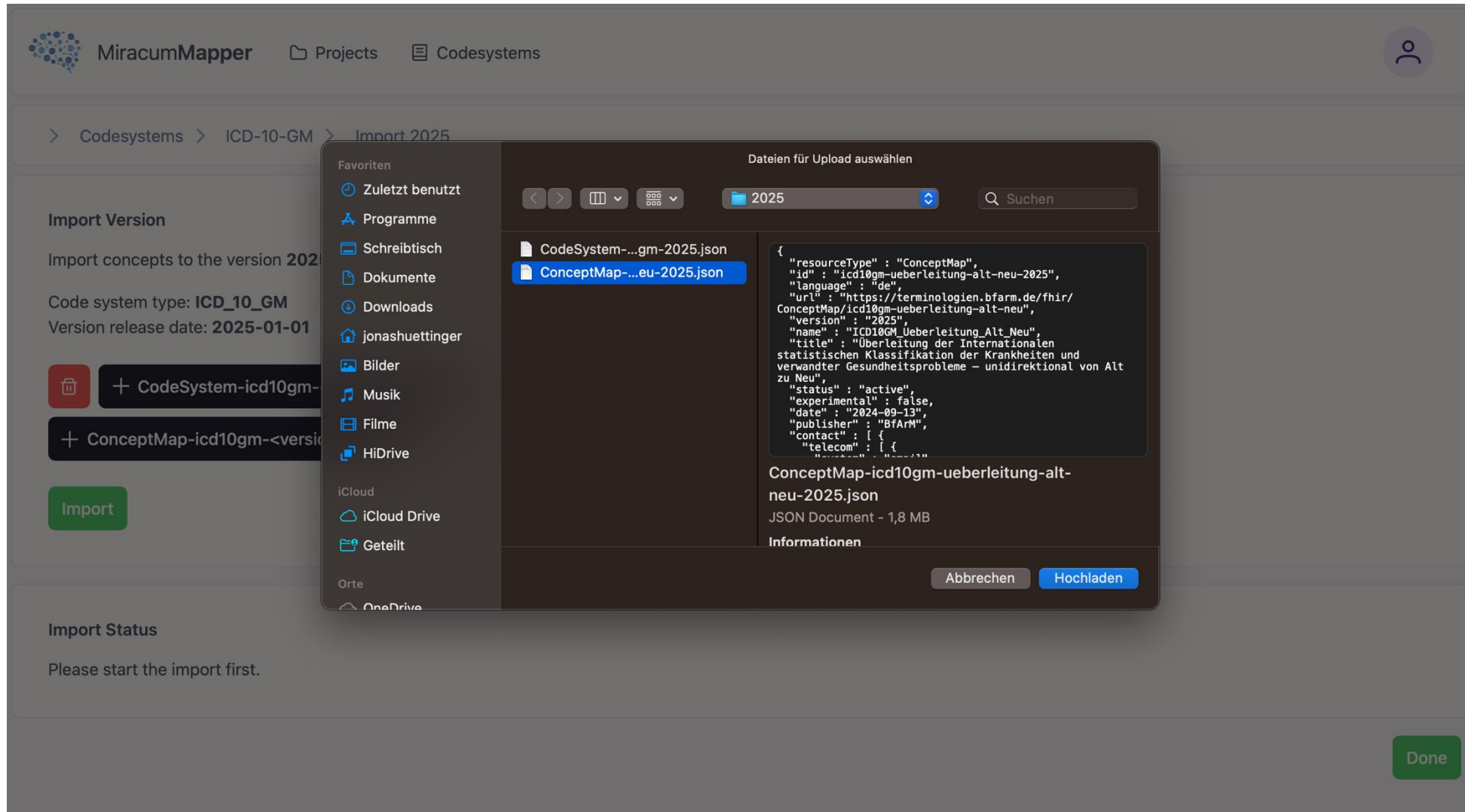
No file chosen 

Import

Import Status

Please start the import first.

Done



The screenshot shows the MiracumMapper web application interface. The main navigation bar includes the MiracumMapper logo, 'Projects', and 'Codesystems'. The breadcrumb trail indicates the current location: 'Codesystems > ICD-10-GM > Import 2025'.

The 'Import Version' section is active, showing the 'Import concepts to the version 2025' step. The 'Code system type' is set to 'ICD_10_GM' and the 'Version release date' is '2025-01-01'. There are buttons to '+ CodeSystem-icd10gm-' and '+ ConceptMap-icd10gm-<version>'. A green 'Import' button is visible.

A file selection dialog titled 'Dateien für Upload auswählen' is open. It shows a list of files in the '2025' folder, including 'CodeSystem-...gm-2025.json' and 'ConceptMap-...eu-2025.json'. The 'ConceptMap-...eu-2025.json' file is selected. The dialog also displays a preview of the JSON content for the selected file, showing details like 'resourceType', 'id', 'language', 'url', 'version', 'name', 'title', 'status', 'experimental', 'date', 'publisher', 'contact', and 'telecom'. The file is identified as 'ConceptMap-icd10gm-ueberleitung-alt-neu-2025.json' and is a 'JSON Document - 1,8 MB'. The dialog has 'Abbrechen' and 'Hochladen' buttons.

The 'Import Status' section at the bottom indicates 'Please start the import first.' and has a green 'Done' button.

 **MiracumMapper** Projects Codesystems 

> Codesystems > ICD-10-GM > Import 2025

Import Version

Import concepts to the version **2025** of code system **ICD-10-GM**.

Code system type: **ICD_10_GM**
Version release date: **2025-01-01**

 **+ CodeSystem-icd10gm-<version>.json *** CodeSystem-icd10gm-2025.json ⓘ

 **+ ConceptMap-icd10gm-<version>.json *** ConceptMap-icd10gm-ueberleitung-alt-neu-2025.json ⓘ

Import

Import Status

Please start the import first.

Done

 **MiracumMapper** Projects Codesystems 

> Codesystems > ICD-10-GM > Import 2025

Import Version

Import concepts to the version **2025** of code system **ICD-10-GM**.

Code system type: **ICD_10_GM**
Version release date: **2025-01-01**

Please wait while the file is being uploaded. This may take a while depending on the size of the file.

Import Status

Please start the import first.

Done

 **MiracumMapper** [Projects](#) [Codesystems](#)

✓ **Success**
File uploaded successfully

> Codesystems > ICD-10-GM > Import 2025

Import Version
Import concepts to the version **2025** of code system **ICD-10-GM**.
Code system type: **ICD_10_GM**
Version release date: **2025-01-01**

The file has been uploaded successfully. You can see the import status in the import status panel.

Import Status
Status: **Running**
Errors: **No errors so far**
Progress:

12%

Done

 **MiracumMapper** Projects Codesystems 

> Codesystems > ICD-10-GM > Import 2025

Import Version

Import concepts to the version **2025** of code system **ICD-10-GM**.

Code system type: **ICD_10_GM**
Version release date: **2025-01-01**

The file has been uploaded successfully. You can see the import status in the import status panel.

Import Status

Status: **Running**
Errors: **No errors so far**
Progress:

75%

Done

 **MiracumMapper** [Projects](#) [Codesystems](#)

✓ **Success**
Codesystem version imported successfully

> Codesystems > ICD-10-GM > Import 2025

Import Version
Import concepts to the version **2025** of code system **ICD-10-GM**.
Code system type: **ICD_10_GM**
Version release date: **2025-01-01**

The file has been uploaded successfully. You can see the import status in the import status panel.

Import Status
Status: **Finished**
Errors: **No errors so far**
Progress:

100%

Done

 MiracumMapper

Projects

Codesystems



Codesystems

ICD-10-GM

ICD-10-GM - Versions

Edit codesystem

+ Add

Edit

Delete

Import concepts

	Name	Release Date	Imported	In use
☐	2024	2024-01-01	No	No
☐	2025	2025-01-01	Yes	No

 **MiracumMapper**  Projects  Codesystems 

> Projects

Project Overview

Sort by Name 

+ Add Project

Labor-Codesystem zu LOINC

1.0.0

Projekt mit 1:1 Mappings vom Labor-Codesystem zu LOINC

20 

<< < 1 to 1 of 1 > >>

 MiracumMapper

Projects

Codesystems



Projects

Labor-Codesystem zu LOINC

Mapping Table

Concept Browser

Start Migration

Edit project

+ Add

Delete

Keyword Search

Table Options

	Source Labor-Codesystem			Target LOINC			Status	Equivalence	Comment
	Status	Code	Meaning	Status	Code	Meaning			
	Any	Search	Search	Any	Search	Search	Any	Any	Search
☐	Active	1GLC	Glucose im Vollblut	Active	13453-6	Glucose mean value Mean Glucose Bld gHb Est-aCnc	Active	Equivalent	
☐	Active	80HDRNA	Hepatitis D-RNA	Active	85512-2	Hepatitis D virus RNA [Units/volume] (viral load) in Serum by NAA with probe detection HDV RNA Ser NAA+probe-aCnc	Active	Related To	
☐	Active	4FOLS	Folsäure	Active	2282-2	Folate [Mass/volume] in Blood Folate Bld-mCnc	Active	Equivalent	

<< < 1 > >>

Showing 1 to 3 of 3 mappings

10

 MiracumMapper

Projects

Codesystems



Projects

Labor-Codesystem zu LOINC

Start Migration

Start Migration

For each source and target code system, the available newer versions are shown here. You can select a version to migrate to.

Source

Codesystem: Labor-Codesystem

Current Version: 2024 (2024-01-01)

Version Name	Release Date	Imported
No newer versions available		

Target

Codesystem: LOINC

Current Version: 2.77 (2024-02-27)

Version Name	Release Date	Imported
2.78	2024-08-06	Yes

Migrate

Cancel

 **MiracumMapper**

Projects

Codesystems

✓ Success

Migration started successfully



> Projects > Labor-Codesystem zu LOINC

Migration

Concept Browser

✓ Finish Migration

✗ Cancel Migration

Edit project

There is currently a migration in progress for this project:

Target

 LOINC 2.77 (2024-02-27) → 2.78 (2024-08-06)

All mappings that use concepts that have been changed or deleted in the new version must be reviewed. 

Concepts that need a review

✓ Save Changes

1 Deprecated Concepts

2 Meaning Changed

13453-6

Active



Migration

Concept Browser

Finish Migration

Cancel Migration

Edit project

There is currently a migration in progress for this project:

Target LOINC 2.77 (2024-02-27) → 2.78 (2024-08-06)

All mappings that use concepts that have been changed or deleted in the new version must be reviewed. ⓘ

Concepts that need a review

Save Changes

1 Deprecated Concepts

2 Meaning Changed

13453-6 Active

Old meaning: Glucose mean value | Mean Glucose Bld gHb Est-aCnc
New meaning: Deprecated Glucose mean value | Deprecated Mean Glucose Bld gHb Est-aCnc

Choose review option for all mappings: Review later / Decide individually for each mapping

Suggested new concepts:

- 27353-2 Active Comment: if units are mg/dL ⓘ Select
- 53553-4 Active Comment: if units are mmol/L ⓘ Select

Source Labor-Codesystem			Target LOINC			Status	Equivalence	Comment	Review option	New concept
Status	Code	Meaning	Status	Code	Meaning					
Active	1GLC	Glucose im Vollblut	Active	13453-6	Glucose mean value Mean Glucose Bld gHb Est-aCnc	Active	Equivalent		Use default option	

Migration

Concept Browser

Finish Migration

Cancel Migration

Edit project

There is currently a migration in progress for this project:

Target

 LOINC 2.77 (2024-02-27) → 2.78 (2024-08-06)

All mappings that use concepts that have been changed or deleted in the new version must be reviewed. ⓘ

Concepts that need a review

Save Changes

1 Deprecated Concepts

2 Meaning Changed

13453-6 Active

Old meaning: Glucose mean value | Mean Glucose Bld gHb Est-aCnc
New meaning: Deprecated Glucose mean value | Deprecated Mean Glucose Bld gHb Est-aCnc

Choose review option for all mappings: Review later / Decide individually for each mapping

Suggested new concepts:

27353-2 Active

 Comment: if ur

53553-4 Active

 Comment: if ur

Review later / Decide individually for each mapping

Keep as deprecated

Replace concept

Delete mappings

Source Labor-Codesystem

Target LOINC

Status	Code	Meaning	Status	Code	Meaning	Status	Equivalence	Comment	Review option	New concept
Active	1GLC	Glucose im Vollblut	Active	13453-6	Glucose mean value Mean Glucose Bld gHb Est-aCnc	Active	Equivalent		Use default option	

Migration

Concept Browser

Finish Migration

Cancel Migration

Edit project

There is currently a migration in progress for this project:

Target LOINC 2.77 (2024-02-27) → 2.78 (2024-08-06)

All mappings that use concepts that have been changed or deleted in the new version must be reviewed. ⓘ

Concepts that need a review

Save Changes

1 Deprecated Concepts

2 Meaning Changed

13453-6 Active

Old meaning: Glucose mean value | Mean Glucose Bld gHb Est-aCnc

New meaning: Deprecated Glucose mean value | Deprecated Mean Glucose Bld gHb Est-aCnc

Choose review option for all mappings: Review later / Decide individually for each mapping

Suggested new concepts:

27353-2 Active Comment: if units are mg/dL ⓘ Select

53553-4 Active Comment: if units are mmol/L ⓘ Meaning: Glucose mean value [Mass/volume] in Blood Estimated from glycated hemoglobin | Est. average glucose Bld gHb Est-mCnc

Source Labor-Codesystem

Target LOINC

Status	Code	Meaning	Status	Code	Meaning	Status	Equivalence	Comment	Review option	New concept
Active	1GLC	Glucose im Vollblut	Active	13453-6	Glucose mean value Mean Glucose Bld gHb Est-aCnc	Active	Equivalent		Use default option	

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There is currently a migration in progress for this project:

Target LOINC 2.77 (2024-02-27) → 2.78 (2024-08-06)

All mappings that use concepts that have been changed or deleted in the new version must be reviewed. ⓘ

Concepts that need a review

✓ Save Changes

- 1 Deprecated Concepts
- 2 Meaning Changed

13453-6 Active

Old meaning: Glucose mean value | Mean Glucose Bld gHb Est-aCnc
New meaning: Deprecated Glucose mean value | Deprecated Mean Glucose Bld gHb Est-aCnc

Choose review option for all mappings: Replace concept

Suggested new concepts:

- 27353-2 Active Comment: if units are mg/dL ⓘ Select
- 53553-4 Active Comment: if units are mmol/L ⓘ Select

New concept: 27353-2

Source Labor-Codesystem			Target LOINC			Status	Equivalence	Comment	Review option	New concept
Status	Code	Meaning	Status	Code	Meaning					
Active	1GLC	Glucose im Vollblut	Active	13453-6	Glucose mean value Mean Glucose Bld gHb Est-aCnc	Active	Equivalent		Use default option	

 MiracumMapper

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Edit project

There is currently a migration in progress for this project:

Target

LOINC

2.77 (2024-02-27)

→

2.78 (2024-08-06)

All mappings that use concepts that have been changed or deleted in the new version must be reviewed. 

Concepts that need a review

Save Changes

1 Deprecated Concepts

2 Meaning Changed

85512-2

Active

^

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Edit project

There is currently a migration in progress for this project:

Target

LOINC

2.77 (2024-02-27)

→

2.78 (2024-08-06)

All mappings that use concepts that have been changed or deleted in the new version must be reviewed. ⓘ

Concepts that need a review

Save Changes

1 Deprecated Concepts

2 Meaning Changed

85512-2

Active

Old meaning: Hepatitis D virus RNA [Units/volume] (viral load) in Serum by NAA with probe detection | HDV RNA Ser NAA+probe-aCnc

New meaning: Hepatitis D virus RNA [Units/volume] (viral load) in Serum or Plasma by NAA with probe detection | HDV RNA SerPI NAA+probe-aCnc

Choose review option for all mappings: Review later / Decide individually for each mapping

Source

Labor-Codesystem

Target

LOINC

Status	Code	Meaning	Status	Code	Meaning	Status	Equivalence	Comment	Review option	New concept
Active	80HDRNA	Hepatitis D-RNA	Active	85512-2	Hepatitis D virus RNA [Units/volume] (viral load) in Serum by NAA with probe detection HDV RNA Ser NAA+probe-aCnc	Active	Related To		Use default option	

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There is currently a migration in progress for this project:

Target LOINC 2.77 (2024-02-27) → 2.78 (2024-08-06)

All mappings that use concepts that have been changed or deleted in the new version must be reviewed. ⓘ

Concepts that need a review

Save Changes

1 Deprecated Concepts

2 Meaning Changed

85512-2 Active

Old meaning: Hepatitis D virus RNA [Units/volume] (viral load) in Serum by NAA with probe detection | HDV RNA Ser NAA+probe-aCnc
New meaning: Hepatitis D virus RNA [Units/volume] (viral load) in Serum or Plasma by NAA with probe detection | HDV RNA SerPI NAA+probe-aCnc

Choose review option for all mappings: Review later / Decide individually for each mapping

Source Labor-Codesystem

Status	Code	Meaning	Status	Equivalence	Comment	Review option	New concept
Active	80HDRNA	Hepatitis D-RNA	Active	Related To		Use default option	

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Edit project

There is currently a migration in progress for this project:

Target LOINC 2.77 (2024-02-27) → 2.78 (2024-08-06)

All mappings that use concepts that have been changed or deleted in the new version must be reviewed. ⓘ

Concepts that need a review

Save Changes

1 Deprecated Concepts

2 Meaning Changed

85512-2 Active

Old meaning: Hepatitis D virus RNA [Units/volume] (viral load) in Serum by NAA with probe detection | HDV RNA Ser NAA+probe-aCnc
New meaning: Hepatitis D virus RNA [Units/volume] (viral load) in Serum or Plasma by NAA with probe detection | HDV RNA SerPI NAA+probe-aCnc

Choose review option for all mappings: Accept changes

Source Labor-Codesystem

Target LOINC

Status	Code	Meaning	Status	Code	Meaning	Status	Equivalence	Comment	Review option	New concept
Active	80HDRNA	Hepatitis D-RNA	Active	85512-2	Hepatitis D virus RNA [Units/volume] (viral load) in Serum by NAA with probe detection HDV RNA Ser NAA+probe-aCnc	Active	Related To		Use default option	

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Migration

 Concept Browser  Finish Migration  Cancel Migration  Edit project

There is currently a migration in progress for this project:

Target LOINC 2.77 (2024-02-27) → 2.78 (2024-08-06)

All mappings that use concepts that have been changed or deleted in the new version must be reviewed. 

All changes are reviewed. You can finish the migration.

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Keyword Search

Table Options

	Source Labor-Codesystem			Target LOINC			Status	Equivalence	Comment
	Status	Code	Meaning	Status	Code	Meaning			
	Any	Search	Search	Any	Search	Search	Any	Any	Search
☐	Active	1GLC	Glucose im Vollblut	Active	27353-2	Glucose mean value [Mass/volume] in Blood Estimated from glycated hemoglobin Est. average glucose Bld gHb Est-mCnc	Migrated	Equivalent	--- Migrat 07-27: Co mean val Est-aCnc 2] (Gluc [Mass/vol from gly average g ---
☐	Active	80HDRNA	Hepatitis D-RNA	Active	85512-2	Hepatitis D virus RNA [Units/volume] (viral load) in Serum or Plasma by NAA with probe detection HDV RNA SerPI NAA+probe-aCnc	Migrated	Related To	--- Migrat 07-27: Dis [Hepatitis [Units/vol by NAA w RNA Ser [Hepatitis [Units/vol

Live Demo

Migration eines Projekts



+ Add

Delete

Keyword Search

Table Options

Target LOINC									
Meaning ↑↓	Status ↑↓	Code ↑↓	Meaning ↑↓	Status ↑↓	Equivalence ↑↓	Comment ↑↓			
Search	Any	Search	Search	Any	Any	Search			
Glucose im Vollblut	Active	27353-2	Glucose mean value [Mass/volume] in Blood Estimated from glycated hemoglobin Est. average glucose Bld gHb Est-mCnc	Migrated	Equivalent	--- Migrated target LOINC on 2025-07-27: Concept [13453-6] (Glucose mean value Mean Glucose Bld gHb Est-aCnc) was replaced by [27353-2] (Glucose mean value [Mass/volume] in Blood Estimated from glycated hemoglobin Est. average glucose Bld gHb Est-mCnc). ---			
Hepatitis D-RNA	Active	85512-2	Hepatitis D virus RNA [Units/volume] (viral load) in Serum or Plasma by NAA with probe detection HDV RNA SerPI NAA+probe-aCnc	Migrated	Related To	--- Migrated target LOINC on 2025-07-27: Display has changed from [Hepatitis D virus RNA [Units/volume] (viral load) in Serum by NAA with probe detection HDV RNA Ser NAA+probe-aCnc] to [Hepatitis D virus RNA [Units/volume] (viral load) in Serum or Plasma by NAA with probe detection HDV RNA SerPI NAA+probe-aCnc]. ---			
Folsäure	Active	2282-2	Folate [Mass/volume] in Blood Folate Bld-mCnc	Active	Equivalent				

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